Professional Integration Developer

OBJECTIVES

By the end of this course, you should be able to:

01 Design complex integration processes to implement advanced logic and process data

02 Debug and troubleshoot simple integration processes

03 Design and deploy event-based, web service integration processes

04 Translate business scenarios into AtomSphere integration processes

Course Overview

In Professional Integration Developer, we cover the following topics:

- Extensions
- REST
- SOAP
- Process Call
- Business Rules
- Document Caching
- Try/Catch
- Errors & Error Handling
- Web Services

Extensions

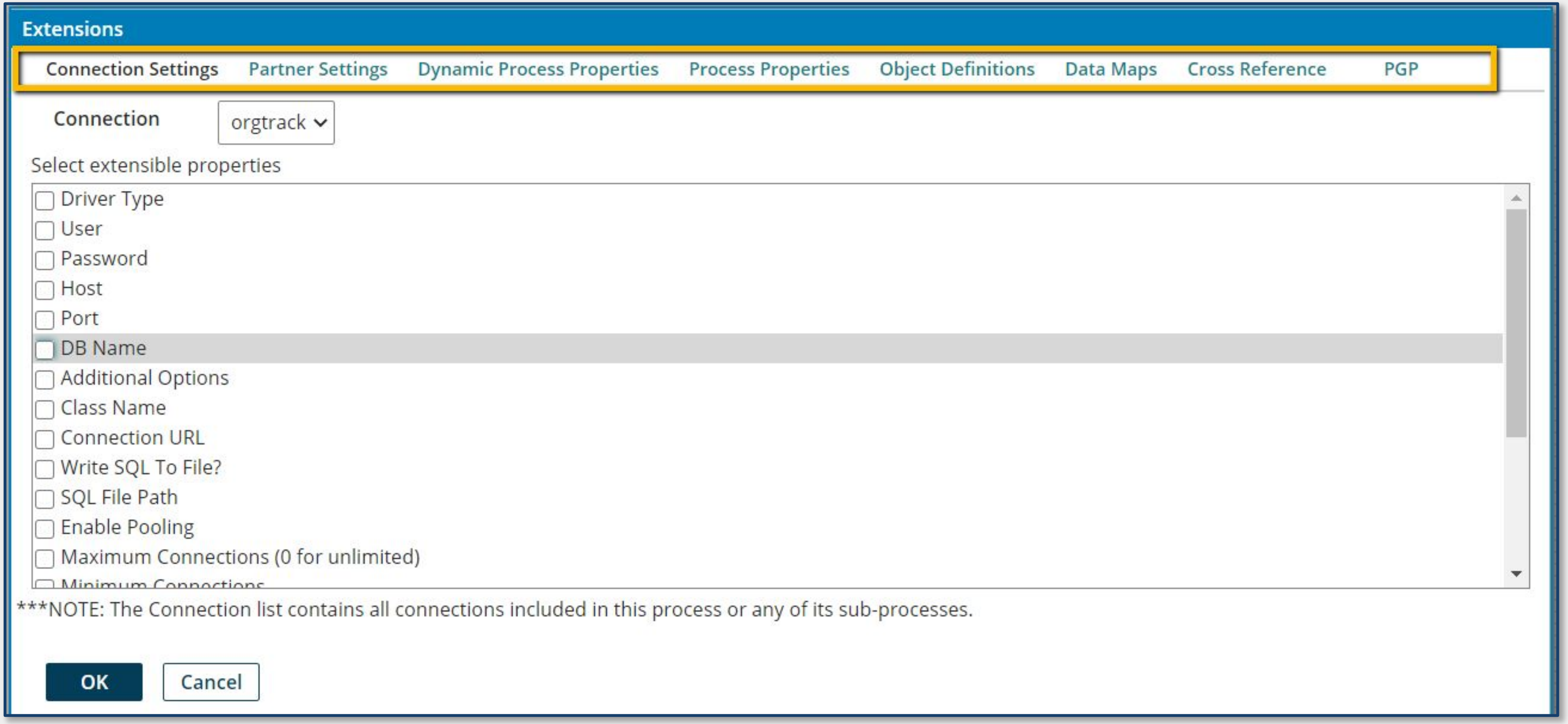
What are extensions?

Allow process configurations after deployment

Configured on the environment level

Configured per process

Types of settings for Extensions

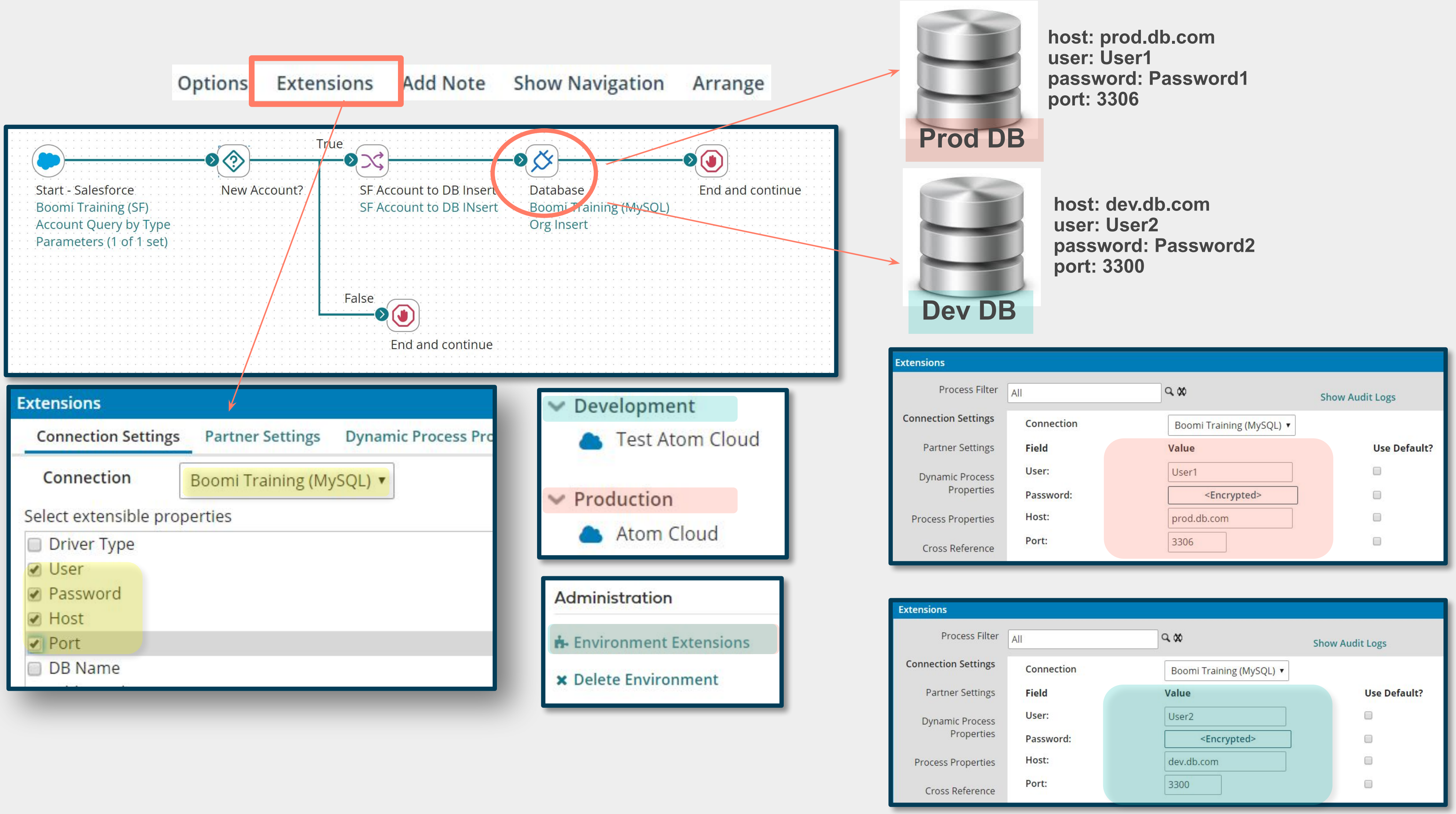


Extensions – Use Cases

1. Environment specific configurations
    - Most common use case
    - Allows configurations to be defined at the environment level instead of hardcoding values

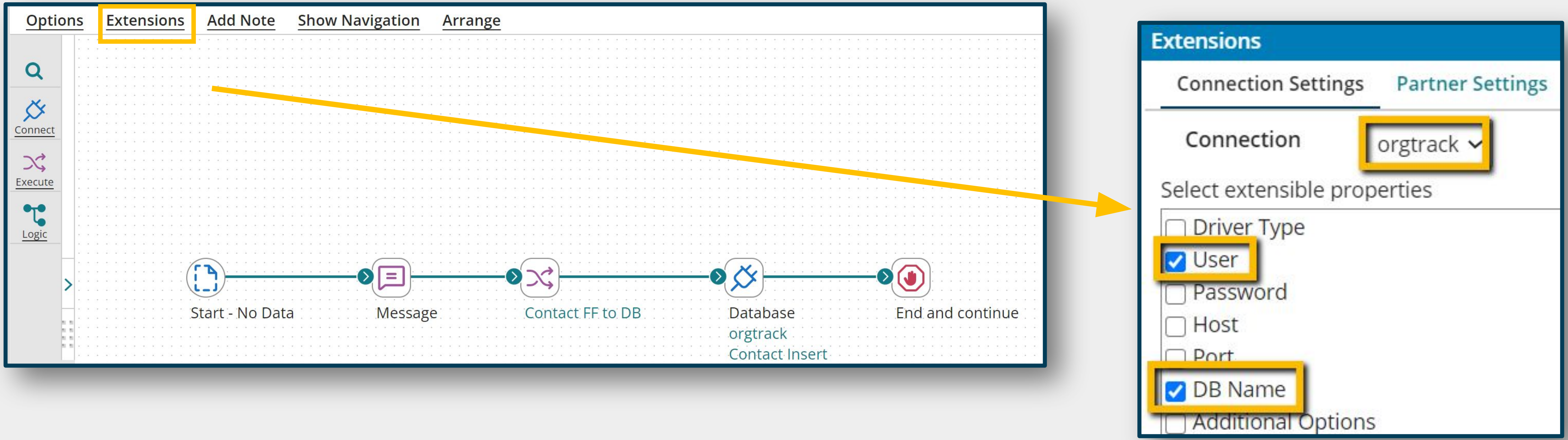
2. Integration Packs
    - Allows integration pack users to customize by setting extension values.

Process Extensions: Use Case



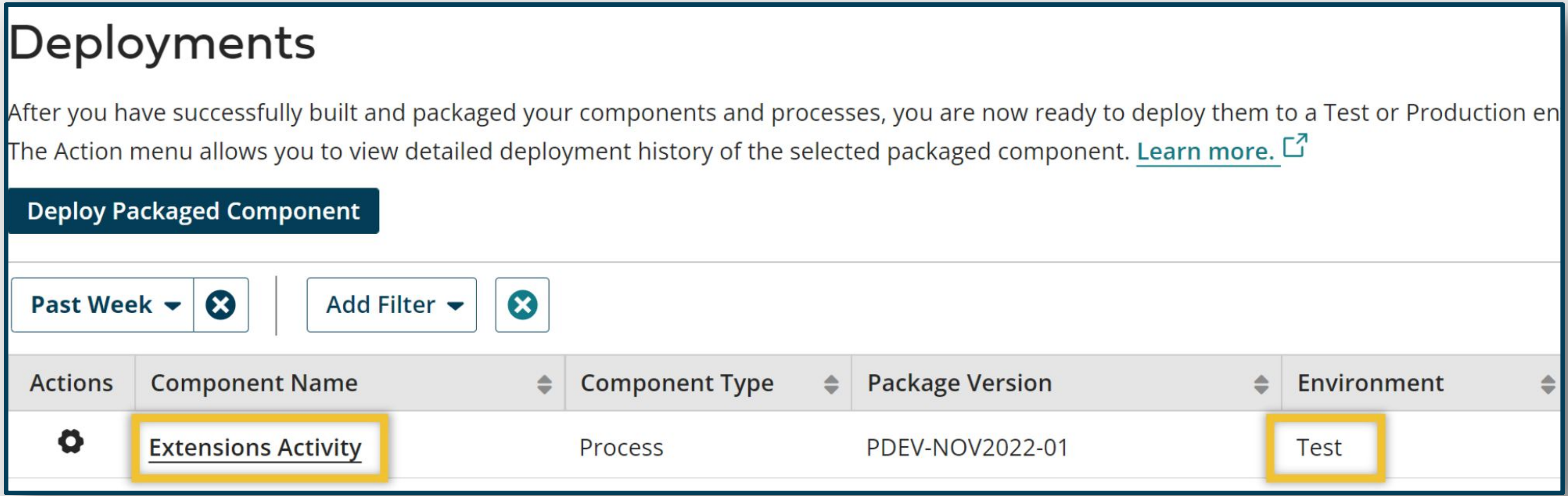
Process Extensions: Setup

1. **Activate extensions and define extensible components within process**



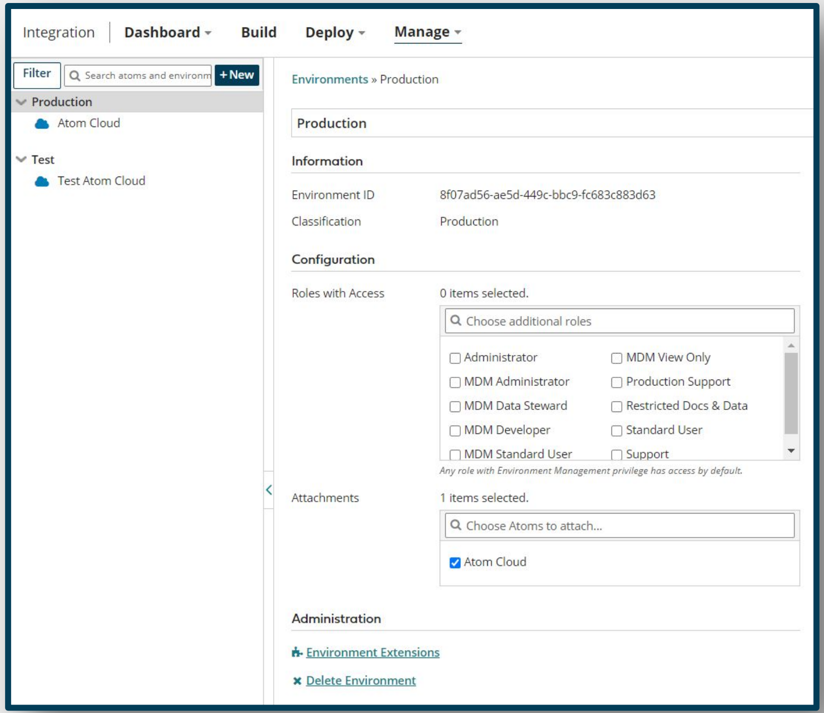
Process Extensions: Setup

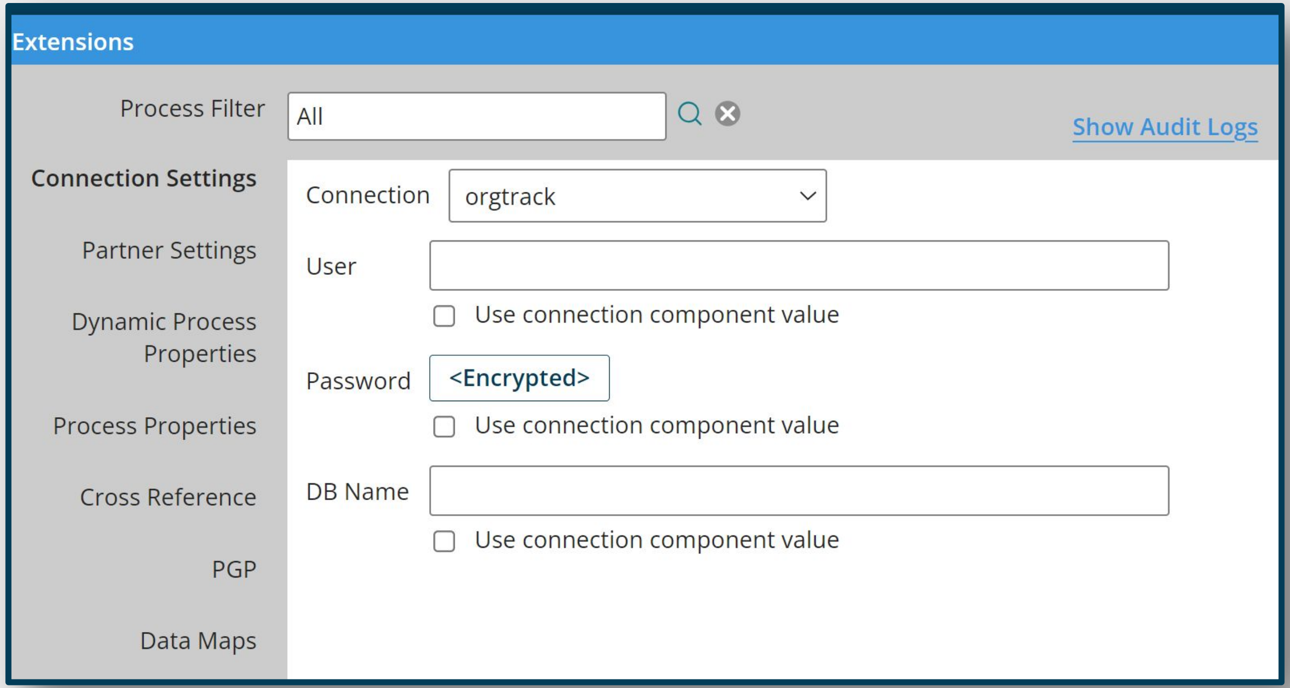
1. Activate extensions and define extensible components within process
2. **Deploy process to the Environment**



Process Extensions: Setup

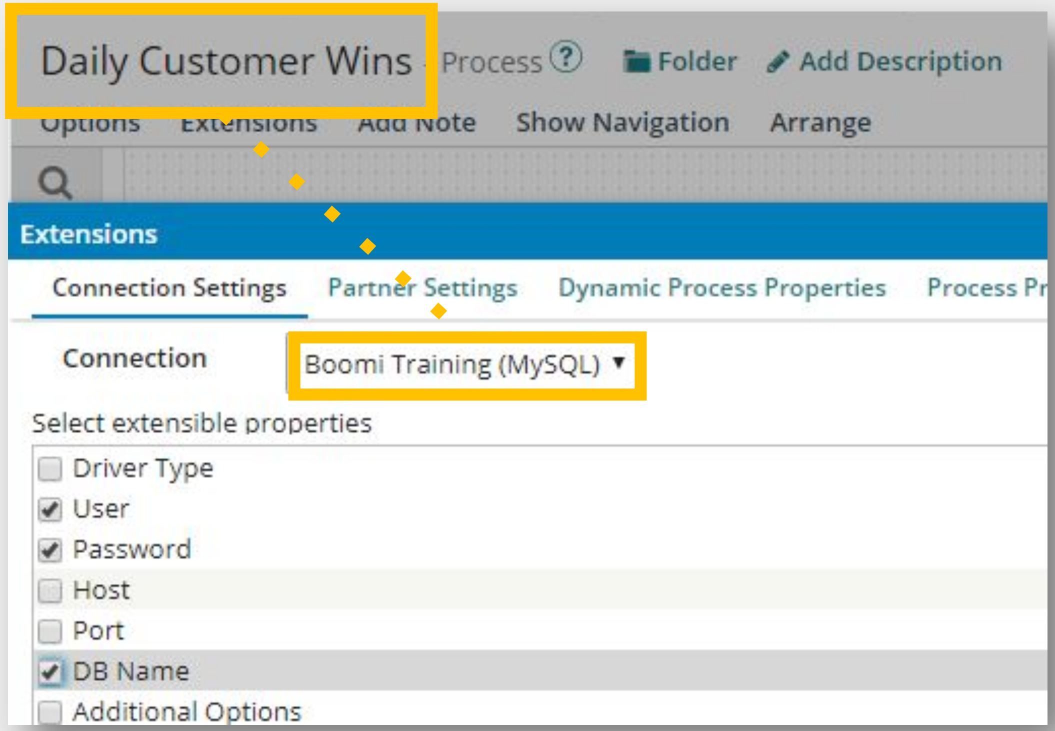
1. Activate extensions and define extensible components within process
2. Deploy process to the Environment
3. **Configure Environment extensions**

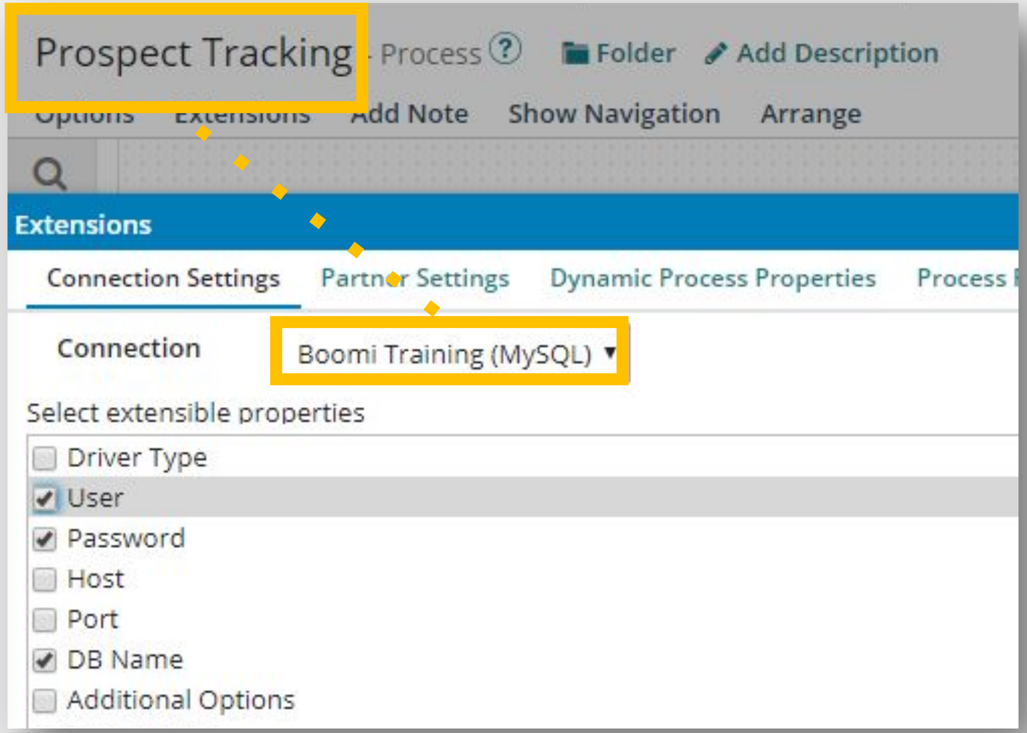




How do Extensions work?

- Extensions are defined per process
- When components (e.g., Connection) are referenced by multiple processes, EACH process must extend that component





Process Extensions in Parent/Child Processes

By Process Call Shape

Child process called are deployed as part of the parent:

- Define extensions in the parent, then extensions are applied to both the parent and child processes
- Define extensions in a child process, then extensions are not applied during execution of the parent process
- Independently deploy the child process, extensions are applied when the child process executes independently of the parent process

By Process Route Shape

Child process are not deployed as part of the parent:

- Extensions defined in the parent process are applied only to the parent process.
- Extensions defined in a child process are applied only to child process and not the parent process or any other child process.

Process Extensions: Recommendations

- Do not connect to the wrong endpoint
- Extend connection changed fields
- Create a placeholder

Extensions are configurable via which of the following

A Build > Create

B Deploy > Attachments

C Manage > Process Reporting

D Manage > Atom Management

When setting up Extensions, which of the following represents the basic 3-step process?

A #1) Deploy Process to Env(s), #2) Configure Env Extensions, #3) Enable Extensions & Define Extensible Components within Process

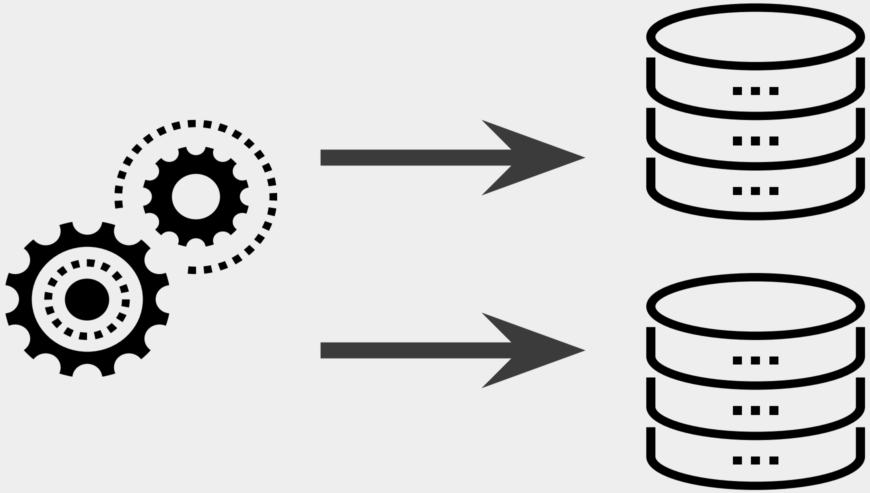
B #1) Configure Env Extensions, #2) Enable Extensions & Define Extensible Components within Process, #3) Deploy Process to Env(s)

C #1) Enable Extensions & Define Extensible Components within Process, #2) Deploy Process to Env(s), #3) Configure Env Extensions

D #1) Enable Extensions & Define Extensible Components within Process, #2) Configure Env Extensions, #3) Deploy Process to Env(s)

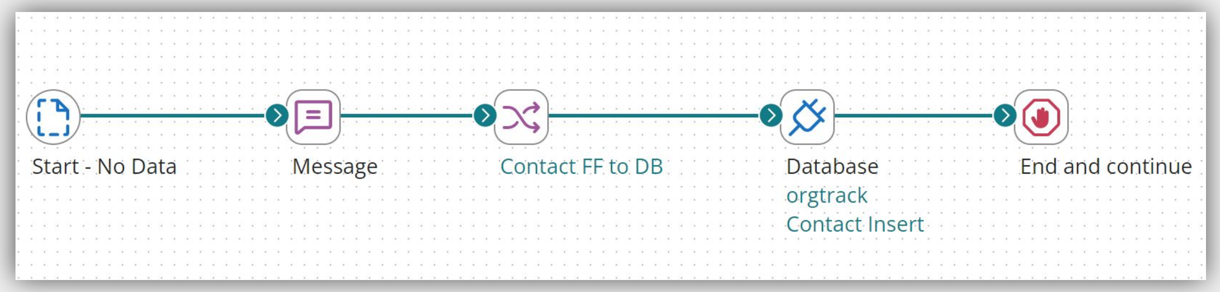
Business Use Case for Extensions Activity

Use Extensions to configure login credentials



Extensions Activity: Overview

- Enable Extensions in the process
- Set the Extension Values in the Build tab
- Test Extension Values
- Package and Deploy the Process to Production and Test Environments
- Set Extension Value for Each Environment
  - DEV: orgtrackDEV
  - PROD: orgtrackPROD



HANDS ON ACTIVITY

Extensions Activity

REST

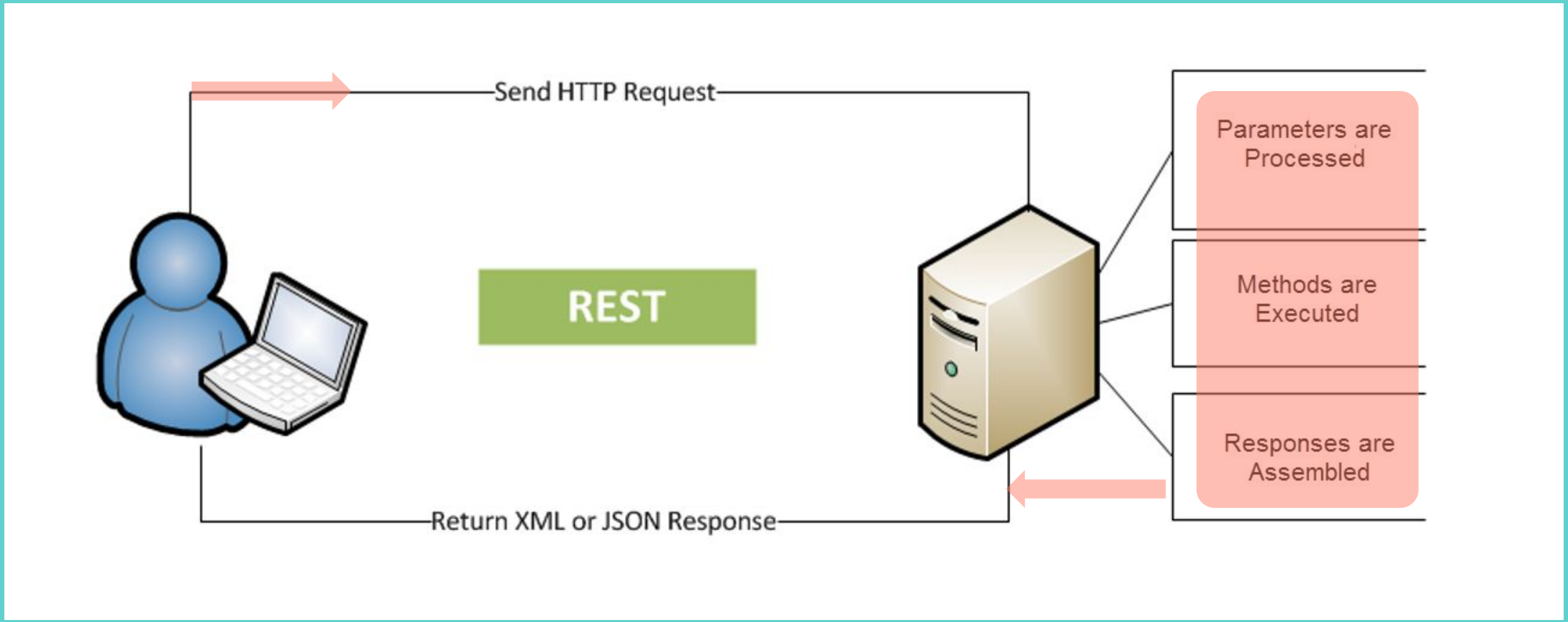
What is REST?

- **R**epresentational **S**tate **T**ransfer (REST)
- Preferred Web Service design model
- Simple architecture style for interacting with networked applications
  - It is simply a URL call
- Platform and software independent
- Easily used in the presence of firewalls
- Depends on a stateless client-server communications protocol (HTTP)
- Does not require XML parsing

A RESTFUL Web Service

- Implemented using HTTP and the principles of REST
- Username, password tokens and API keys are often used for authentication
- A collection of resources, with four design principles:
    - A static base URL (referred to as URI) for the web service
    - Types of internet media data supported:
        - XML, JSON
    - Set of operations supported by the web service:
        - GET, PUT, POST, DELETE
    - The API must be hypertext driven (HTTP)
- No official standard for RESTful web services

RESTful Web Service Diagram



What does REST stand for?

A
Representational State Transfer

B
Returned State Transfer

C
Representational System Tethering

D
Required State Transfer

All RESTful web services follow a specific standard.

A. True

B. False

Which of the following are the response types typically returned by a RESTful web service?

A JavaScript

B XML

C JSON

D PHP

REST Request Formation

- REST requests are formed using a URL-like structure, called a URI
- Uniform Resource Identifier (URI)
  - Base URI
  - Collection URI
  - Element URI

Base URI

http://api.supplies.com /products/123

Collection URI

Element URI

- REST requests are "noun" based and easy to interpret

REST Request Example

- An e-commerce store’s API supplies a list of products by performing an HTTP GET request on this URI:

  http://api.supplies.com/**products**

- The XML response, provided by the server, is:

```
<products>
    <product id=“123“ xlink:href=“http://supplies.com/products/123”/>
    <product id=“456“ xlink:href=“http://supplies.com/products/456”/>
    <product id=“789“ xlink:href=“http://supplies.com/products/789”/>
    <product id=“638“ xlink:href=“http://supplies.com/products/638”/>
</products>
```

REST Request Example

- To GET specific details about product ‘456’, the request might look like:

http://api.supplies.com/products/**456**

- The XML response, provided by the server, is similar to:

```xml
<product>
    <Product-ID>456</Product-ID>
    <Name>Purple widget</Name>
    <Description>A small, right-handed widget</Description>
    <Stock-level>14</Stock-level>
    <UnitCost currency="USD">0.10</UnitCost>
    <Specs xlink:href="http://supplies.com/parts/00345/specification"/>
</product>
```

REST Authentication

- There is no REST web service standard, so differences exist in how a web service authenticates request.
- Types of upfront REST authentication:
  - Basic: Username & Password (Key) required
  - oAuth: Upfront handshake communication with web service to establish Access Token
  - None: If required, API key provided within HTTP URI build request
    - Some open APIs do not require an API key
  - Authentication example within the URI
    - http://api.weather.com/US/PA/Berwyn/?apikey=r6y6snswpu8

REST requests are formed using a concatenated string-like structure, called a(n):

A. URL

B. URI

C. UIT

D. USA

In the following request, which most likely represents the 'base' URI?

http://api.espn.com/v1/sports/football/nfl/teams/?apikey=r6y6snswpu

A api.espn.com/v1/sports/

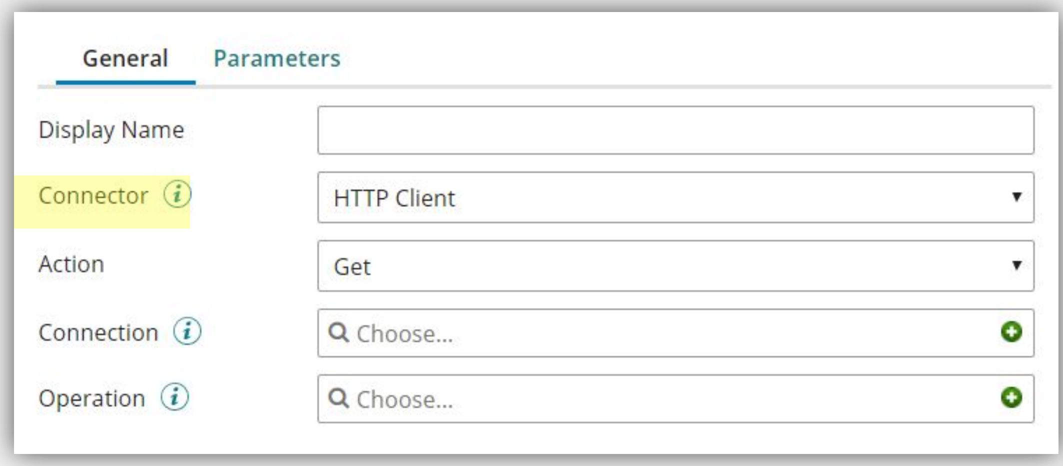
B ?apikey=r6y6snswpu

C api.espn.com

D api.espn.com/v1/sports/football/nfl/teams/

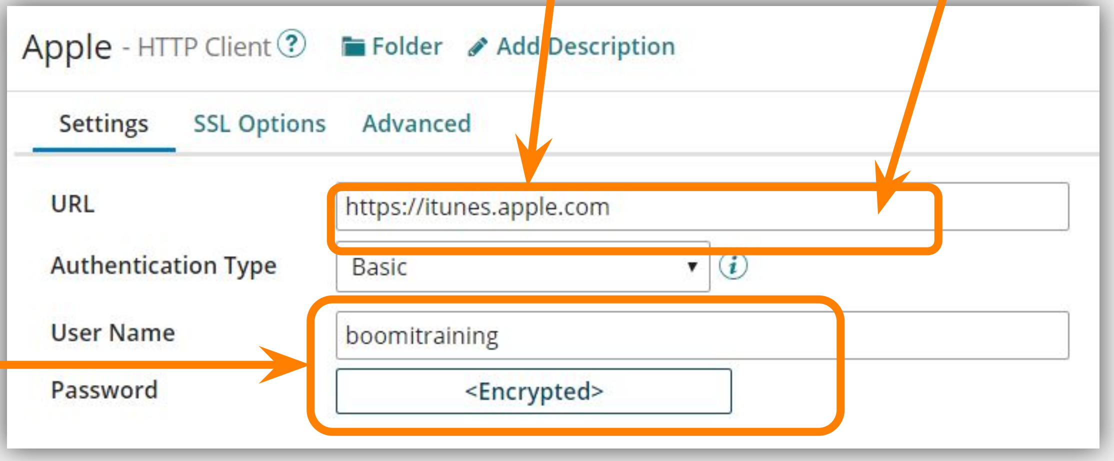
AtomSphere's HTTP Client Connector

Use the HTTP Client Connector to establish REST connectivity



Base URI entered here:

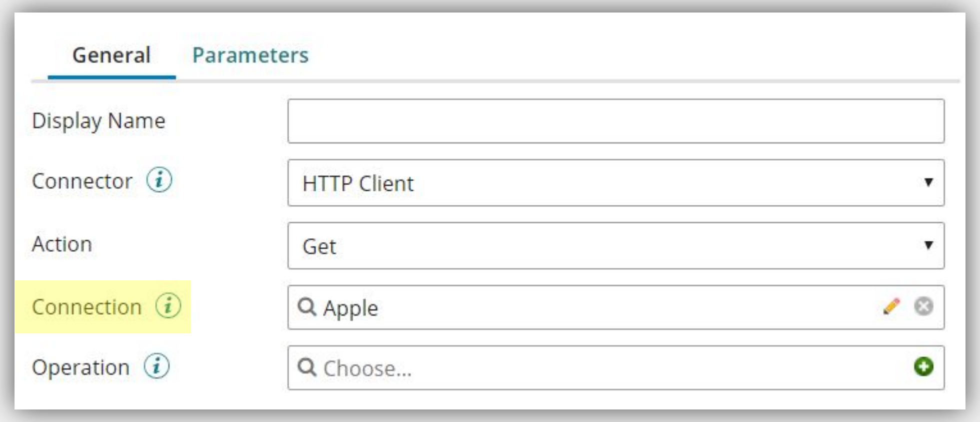
Ending forward slash '/' is ignored

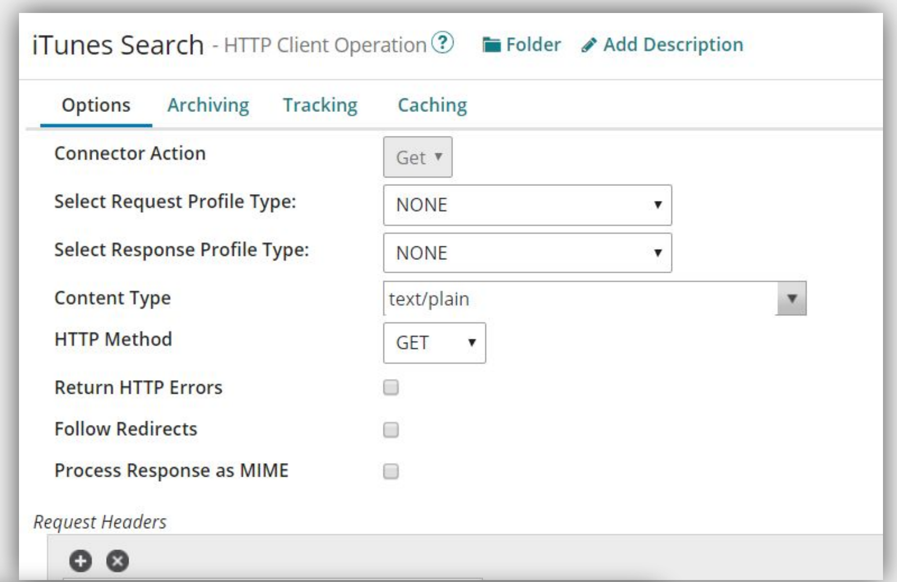


Depending on Auth Type, you enter the credentials received from the REST web service here:

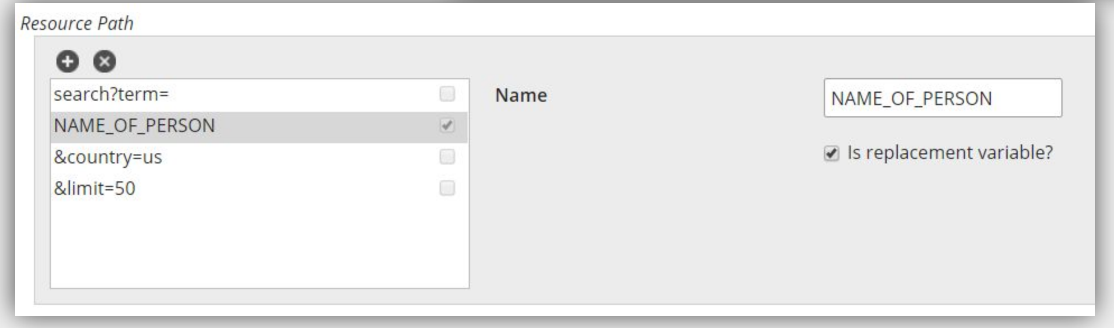
AtomSphere's HTTP Client Connector

Use the HTTP Client Operation to build out Request URI



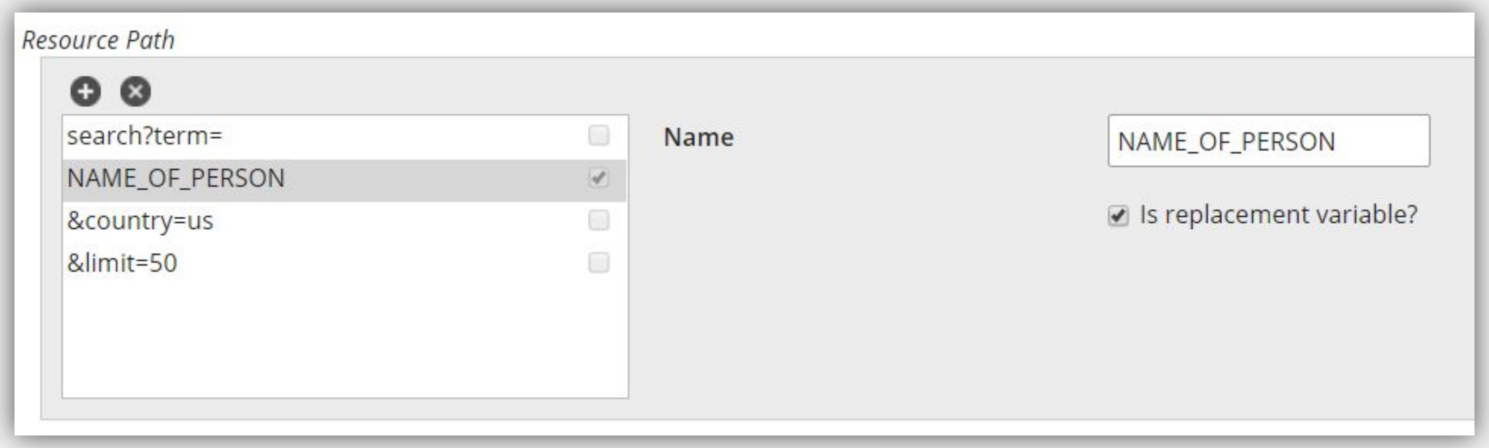


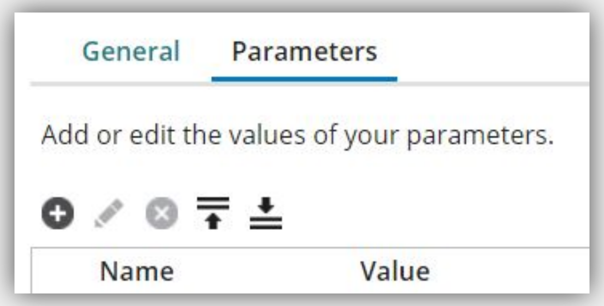
Items to build out the URI request

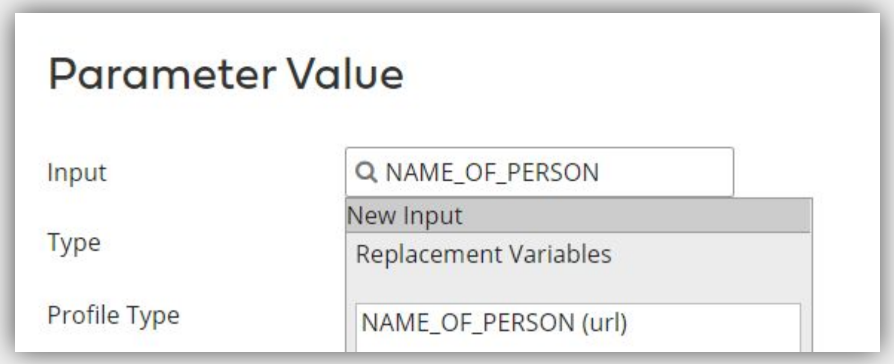


AtomSphere's HTTP Client Connector

Label the Dynamic Resource Path elements as replacement variables defined in the connector's Parameters tab.







Which connector type is used to build out a REST web service connection?

A Web Services Server

B Web Services SOAP Client

C AS2 Shared Server Connector

D HTTP Client

What component of the HTTP Connector contains the Resource Path, representing the elements that build out the URI?

A Connection

B Operation

When setting a Resource Path element as a replacement variable, it is defined and brought into the process via the:

A | Connector Parameter tab

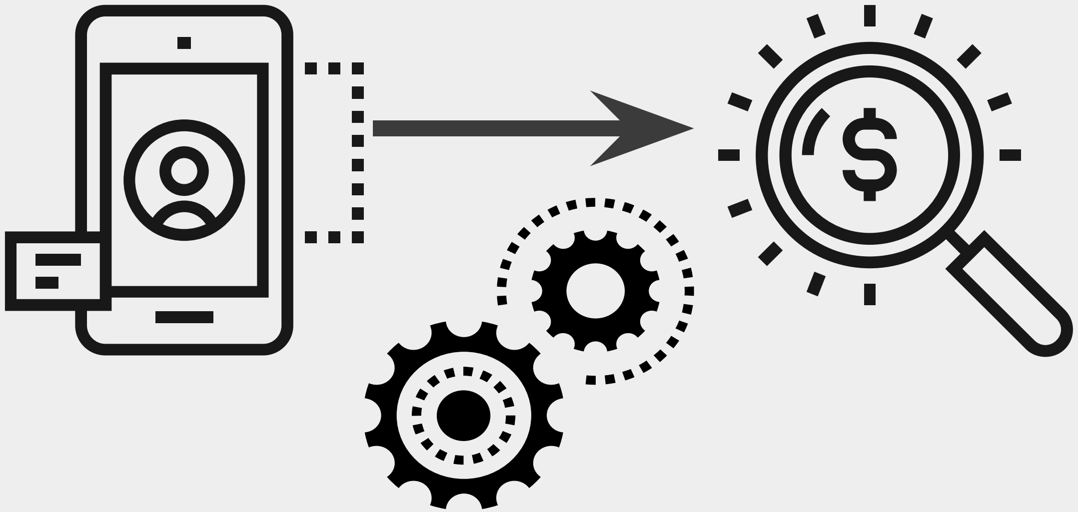
B | Message shape

C | Map

D | Document Property

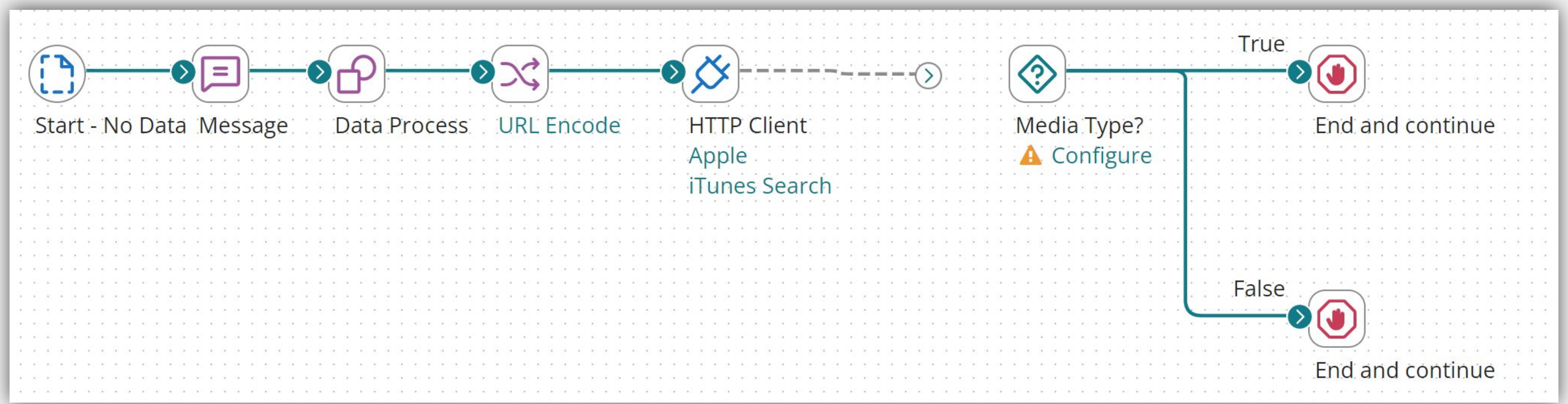
Business Use Case for Activity

- Mobile application developer implementing an interactive application for movie listings and searches.
- User searches for information, results are displayed in the application, and the user can rent or buy the selected title.
- Create and unit test the back-end interface logic for the application.



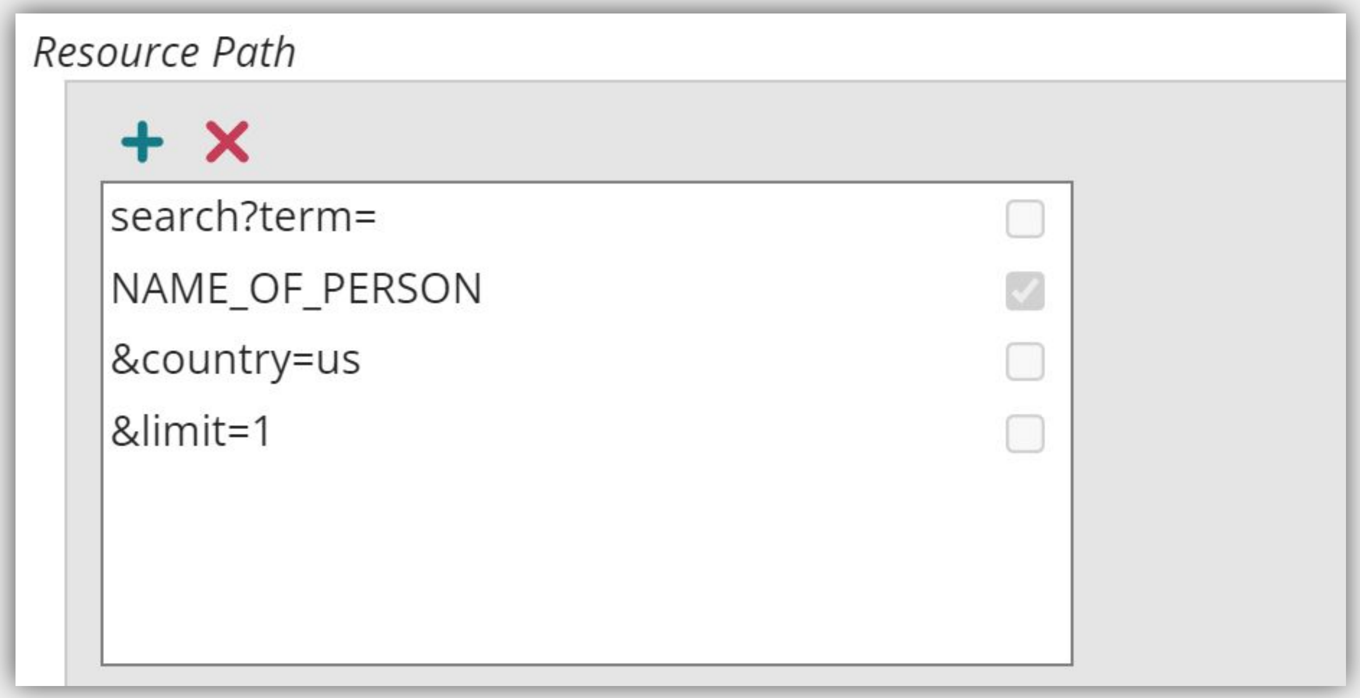
REST Activity: Overview

- Configure the HTTP Client Connector
- Configure the JSON profile, complete the Decision shape, and create output



REST Activity: Overview (section 1)

- Visit the iTunes Search Application to assemble the Request URI. Do not forget the special characters.
- Open the **iTunes Search** operation and scroll down to the Resource Path
- Set the &limit to 1
- Save the output

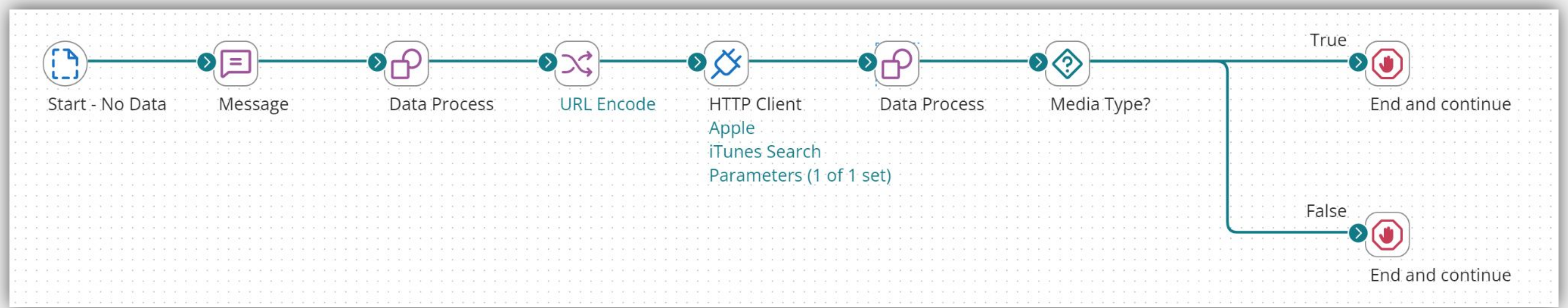


HANDS ON ACTIVITY

Configure the HTTP Connector

Configure and Test the REST Activity

- Upload your new JSON file
- Add a Data Process shape to split the documents
- Configure the Decision shape to test for a feature-movie
- Test your process



Configure and Test the Rest Activity

SOAP

What is SOAP?

- SOAP: “**S**imple **O**bject **A**ccess **P**rotocol”
- Allows applications to communicate with each other
- Simple XML-based communications protocol
  - Information (messages) is usually exchanged over HTTP
- Platform and software independent
- Easily used in the presence of firewalls

A SOAP Web Service

- Unlike a RESTful web service, SOAP web services follow a standardized specification.
- A **WSDL** file describes the SOAP web service
  - Location – the endpoints or ports
  - Methods are available to the client
  - Format of XML data in the request
- Authentication is by username & password tokens

What does SOAP stand for?

A — Simple Object Access Protocol

B — System Object Access Protocol

C — Simple Object Acquire Protocol

D — Simple Object Acquire Protocol

SOAP can be defined as a messaging protocol based on:

A JSON

B CSV

C XML

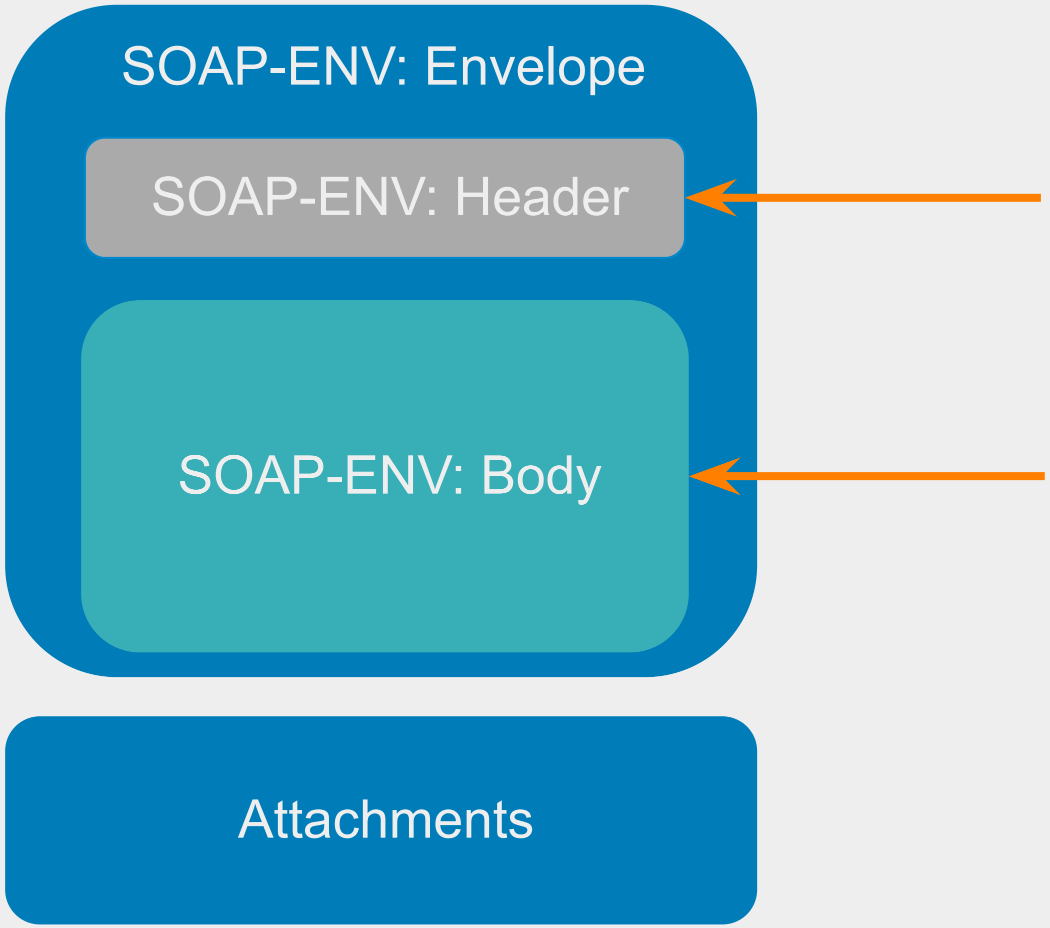
D PHP

The WSDL file describes the methods that are available to the client application, the endpoint (where to call the web service), and the format of the XML that is required in order to call the web service.

A. True

B. False

SOAP Message Structure



Sample SOAP Request

```
POST /InStock HTTP/1.1
Host: www.example.org
Content-Type: application/soap+xml; charset=utf-8
<?xml version="1.0"?>

<soap:Envelope
xmlns:soap="http://www.w3.org/2001/12/soap-envelope"
soap:encodingStyle="http://www.w3.org/2001/12/soap-encoding">

<soap:Header>
  <m:Trans xmlns:m="http://www.w3schools.com/transaction/"
  soap:mustUnderstand="1">234
  </m:Trans>
</soap:Header>

<soap:Body xmlns:m="http://www.example.org/stock">
  <m:GetStockPrice>
    <m:StockName>BockelCorp</m:StockName>
  </m:GetStockPrice>
</soap:Body>

</soap:Envelope>
```

Envelope

Header

Body

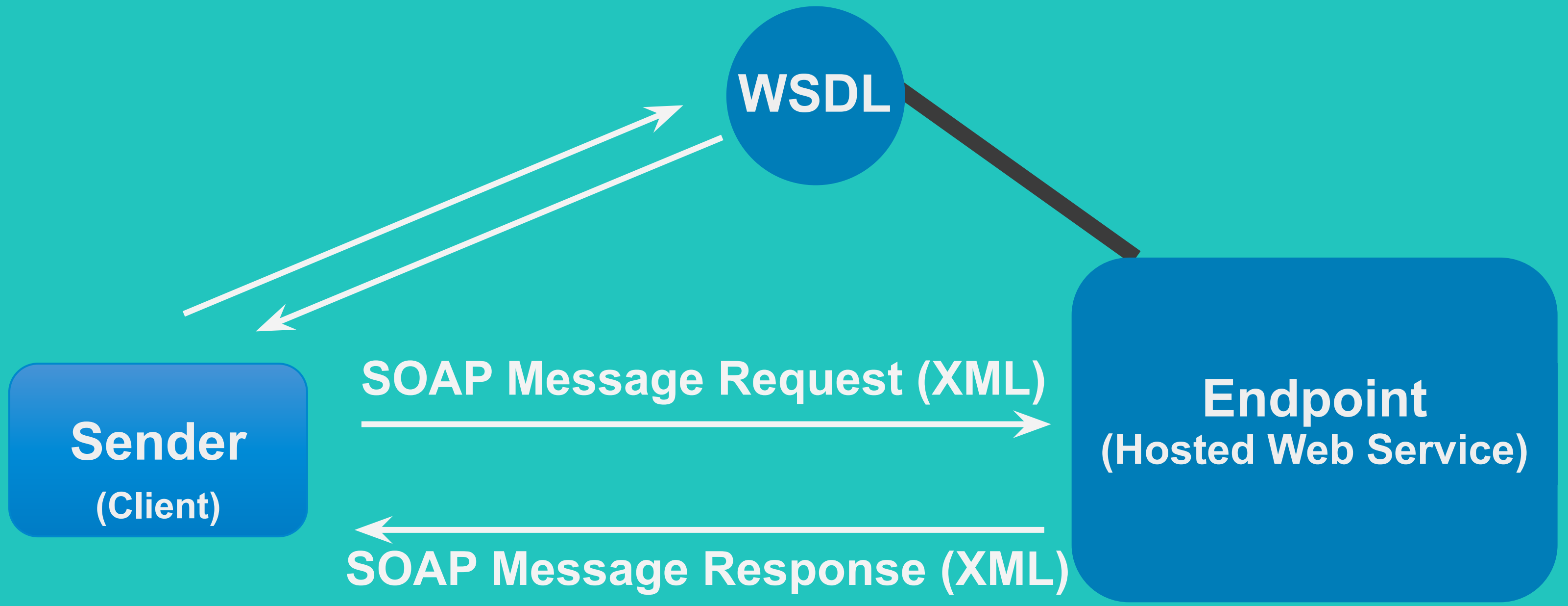
Sample SOAP Response

```
HTTP/1.1 200 OK
Content-Type: application/soap+xml; charset=utf-8
Content-Length: nnn
<?xml version="1.0"?>

<soap:Envelope
xmlns:soap="http://www.w3.org/2001/12/soap-envelope"
soap:encodingStyle="http://www.w3.org/2001/12/soap-encoding">

<soap:Body xmlns:m="http://www.example.org/stock">
  <m:GetStockPriceResponse>
    <m:Price>14.35</m:Price>
  </m:GetStockPriceResponse>
</soap:Body>

</soap:Envelope>
```

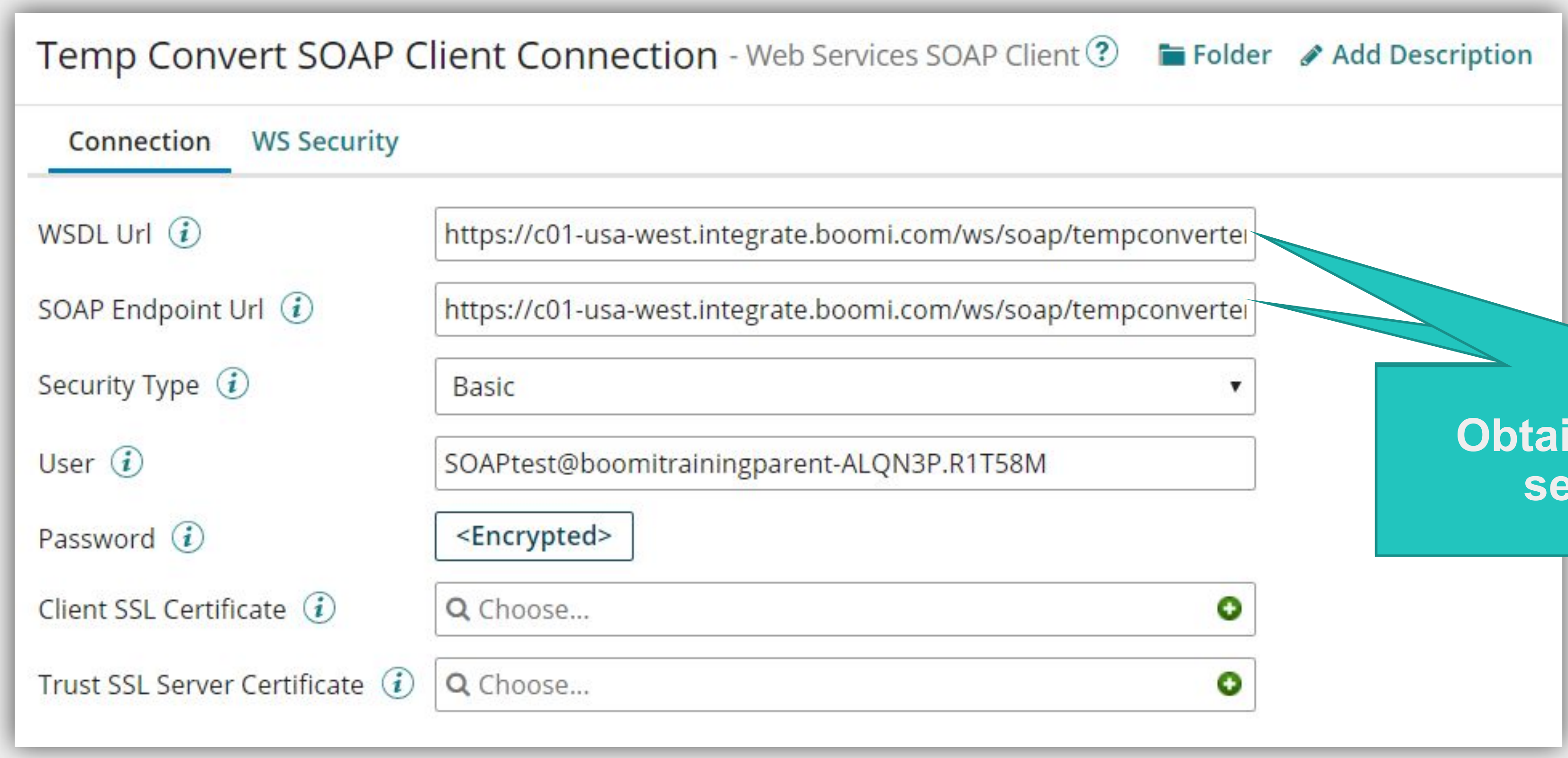
Envelope

Body

A SOAP Web Service

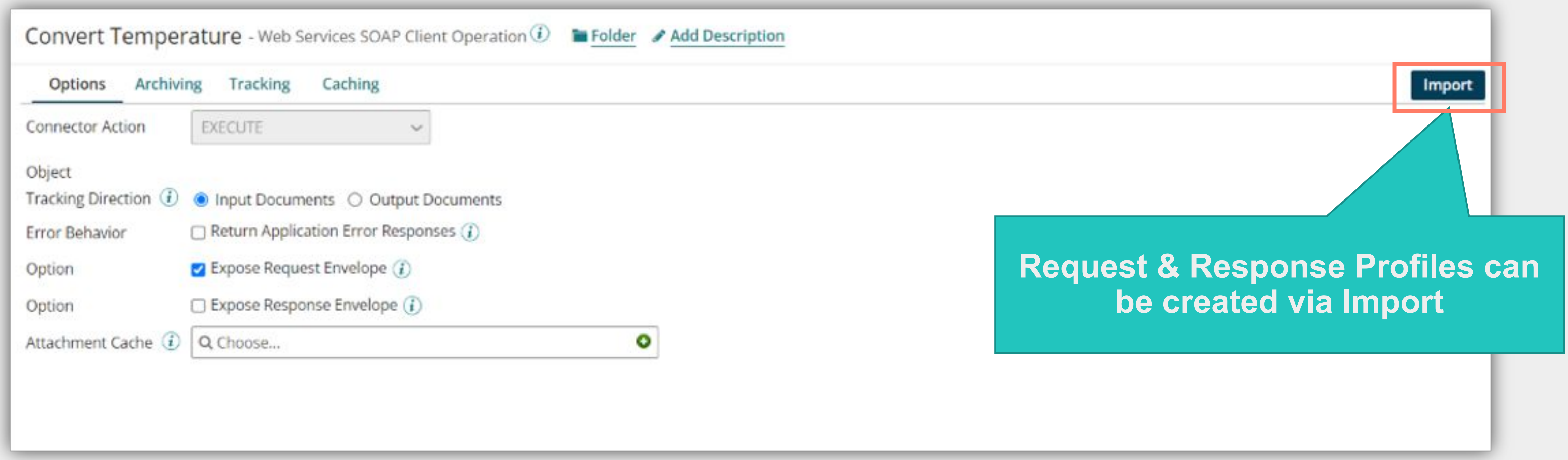


Web Services SOAP Client Connector



Obtain URLs from SOAP web service documentation

Web Services SOAP Client Connector



Which of the following SOAP elements contains the actual SOAP message details intended for the ultimate endpoint?

A Envelope

B Header

C Body

D Attachment

Which connector type is used to build out a SOAP web service connection?

A Web Services Server

B Web Services SOAP Client

C AS2 Shared Server Connector

D HTTP Client

Unlike within the Operation for an HTTP Client (REST) connector, the web services SOAP Client Operation offers the ability to dynamically import both the Request and Response profiles.

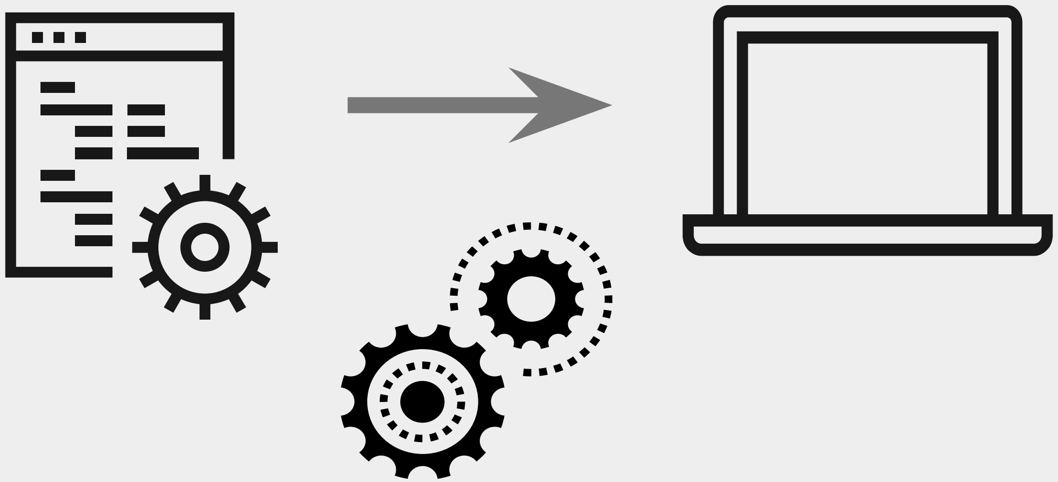
A True

B False

Business Use Case for Activity

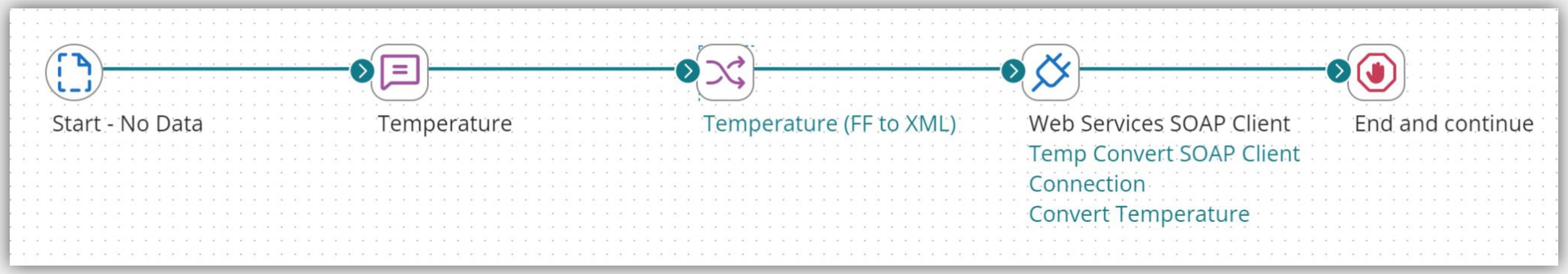
- Web developer implementing a weather application that converts temperature from Celsius to Fahrenheit.

- Create the back-end interface logic between the web service and the front end displaying the converted temperature



SOAP Activity

- Configure your SOAP Client; import request and response profiles
- Enter value (temperature) into Message shape
- Map the value to XML format
- Call Web Service
- Retrieve the new value from temperature conversion



SOAP Activity

Process Call

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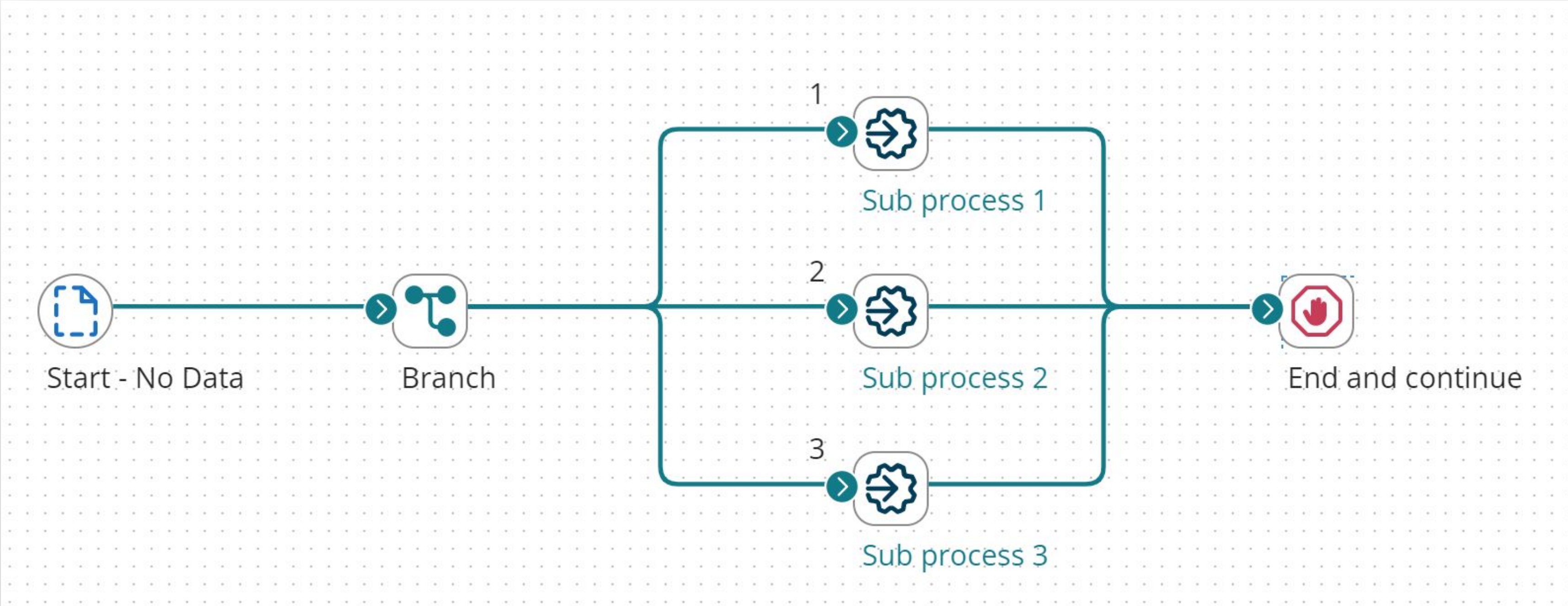
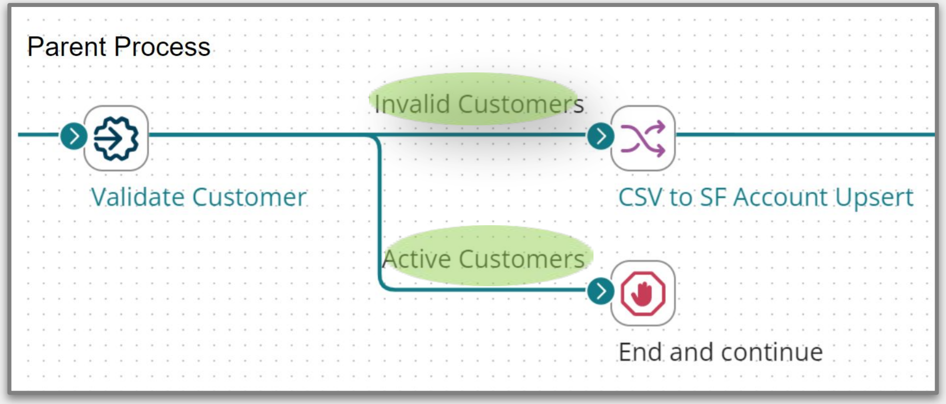
Process Call Shape

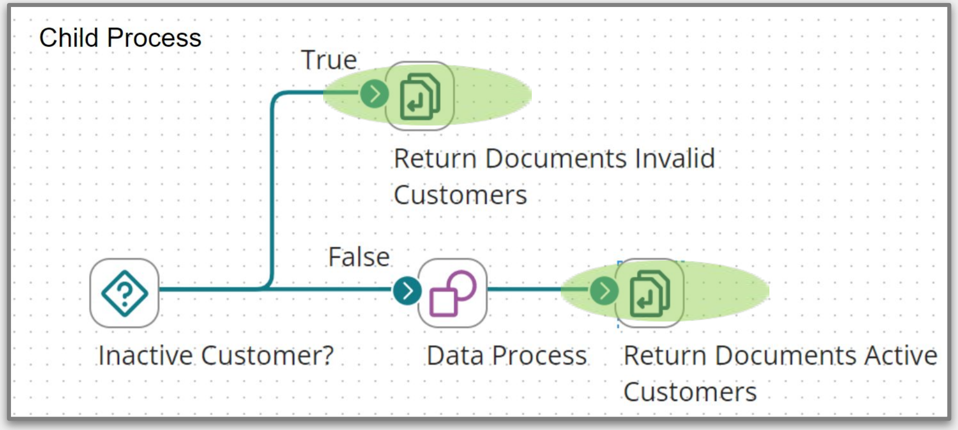
- Executes a sub process (Child) within a Process (Parent)
- Triggers Child Start Shape to run and pass documents into Child workflow
- Organize and simplify complex process architecture



Return Documents Shape

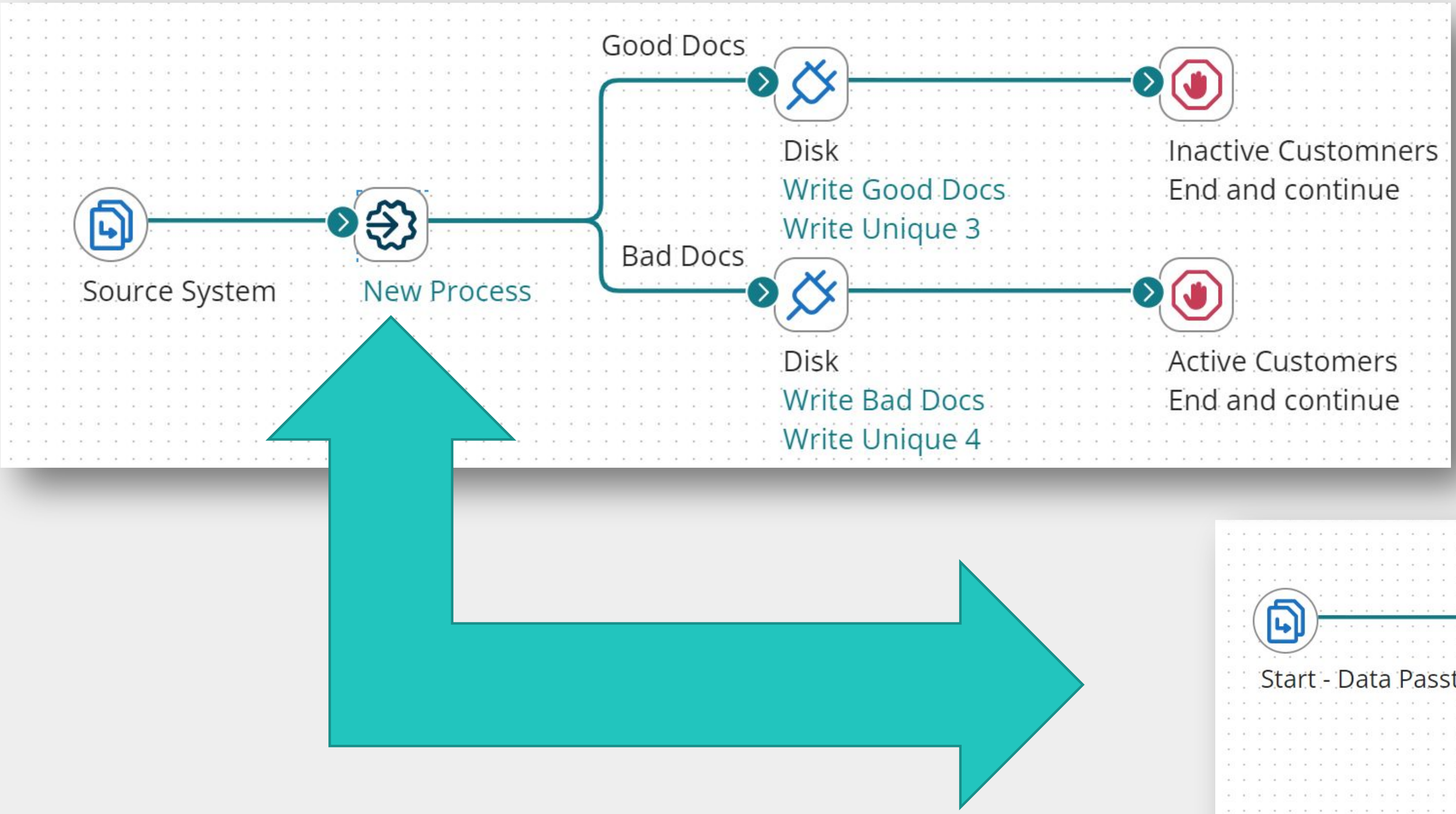
- Returns documents in a Child Process to a Parent Process Call shape
- Waits for all documents to process down paths before executing
- Label creates a corresponding path extending from the Parent Process call



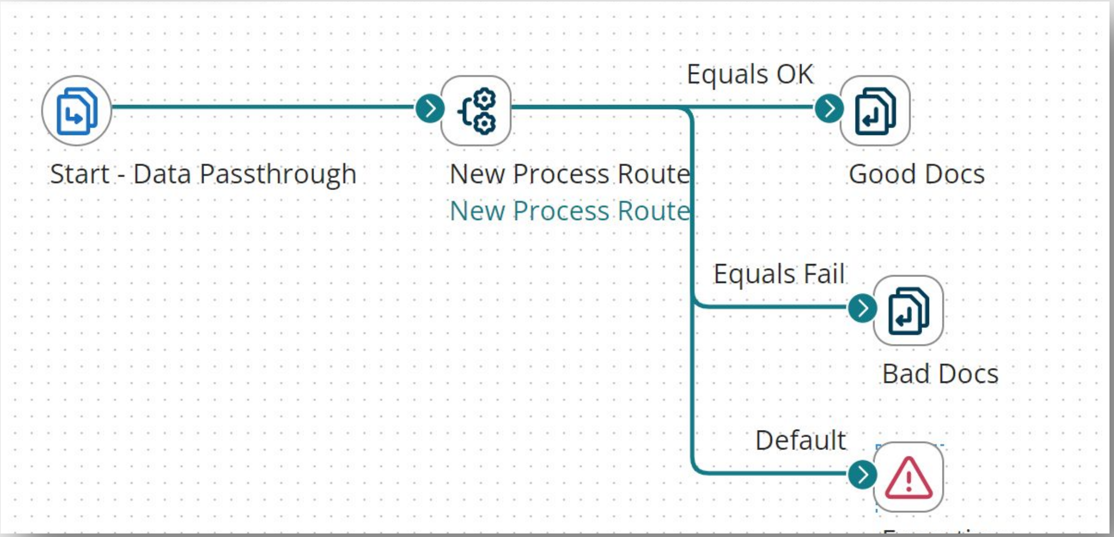


Process Call and Return Documents

Parent Process



Child Process



The Return Documents shape waits for all Documents to process down paths before executing.

A True

B False

Assume a Parent Process utilizes the Process Call shape to trigger a Child Process. Which of the following is ALWAYS true?

A The Child Process must include at least one Return Document Shape

B Only the Parent Process needs to be Packaged and Deployed when putting the Parent/Child Processes into production

C The Parent Process will halt its execution until the Child Process is complete

D The Parent Process will abort if the Child Process results in error

A Parent Process is configured to send a Document to a Child Process (for further processing) through a Process Call shape. How should the Start Shape options be configured at the front-end of the Child Process?

A. Connector

B. Trading Partner

C. Data Passthrough

D. No Data

A Parent Process is configured to trigger a Child Process through a Process Call shape. No Documents are being passed. How should the Start Shape options be configured at the front-end of the Child Process?

A Connector

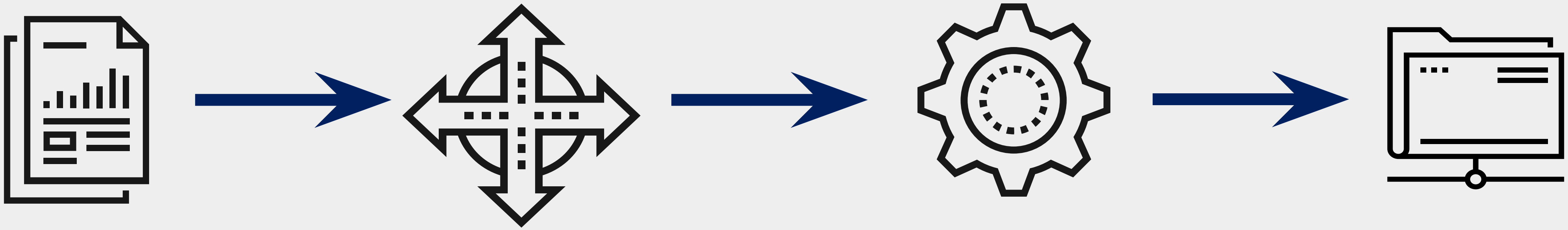
B Trading Partner

C Data Passthrough

D No Data

Business Use Case for Activity

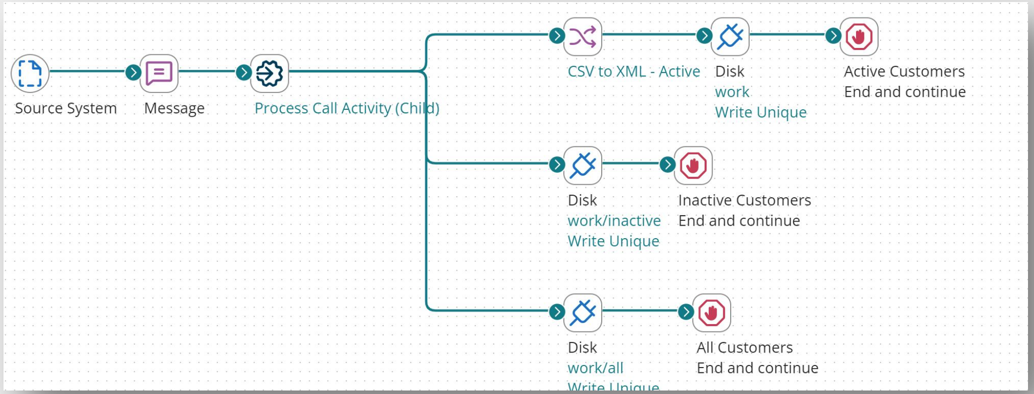
- Process incoming CSV customer records
- Determine the status of the customer records
- Convert active CSV customer records to XML and write them to disk. Write other CSV records to disk



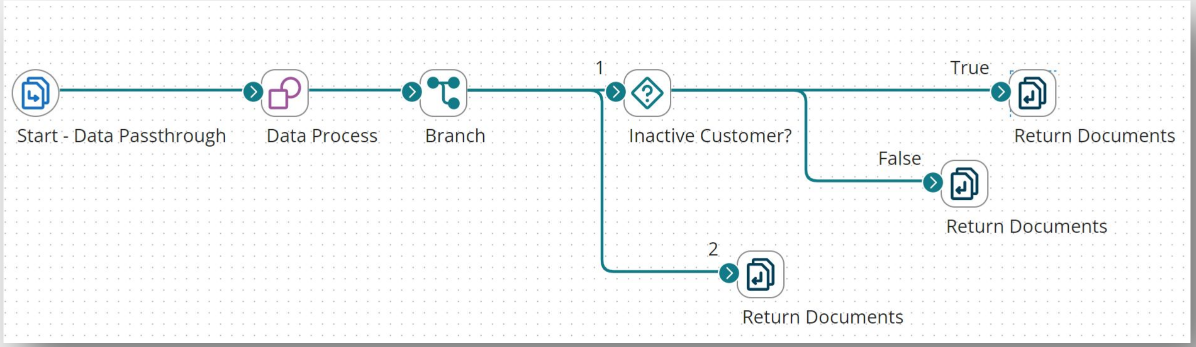
Process Call Activity

- Add a Process Call shape that sends incoming documents to a Child process for evaluation
- In Child process: add and label Return Document shapes based on the status
- In Parent process: connect active, inactive, and all branches to corresponding paths
- Run the Parent process in test mode

PARENT



CHILD

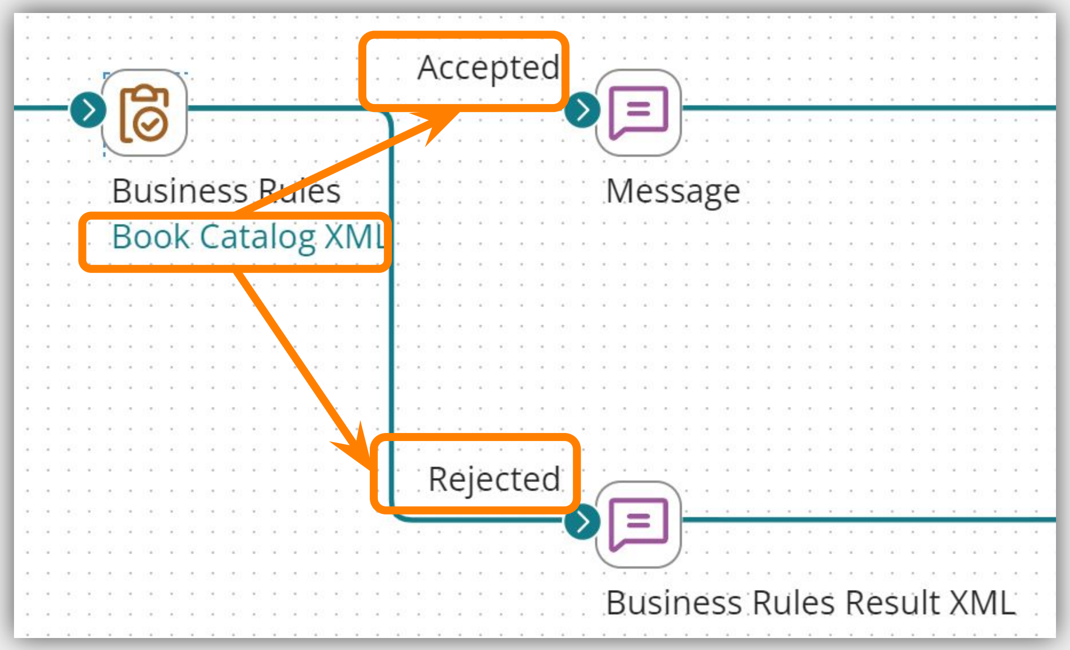


Process Call Activity

Business Rules

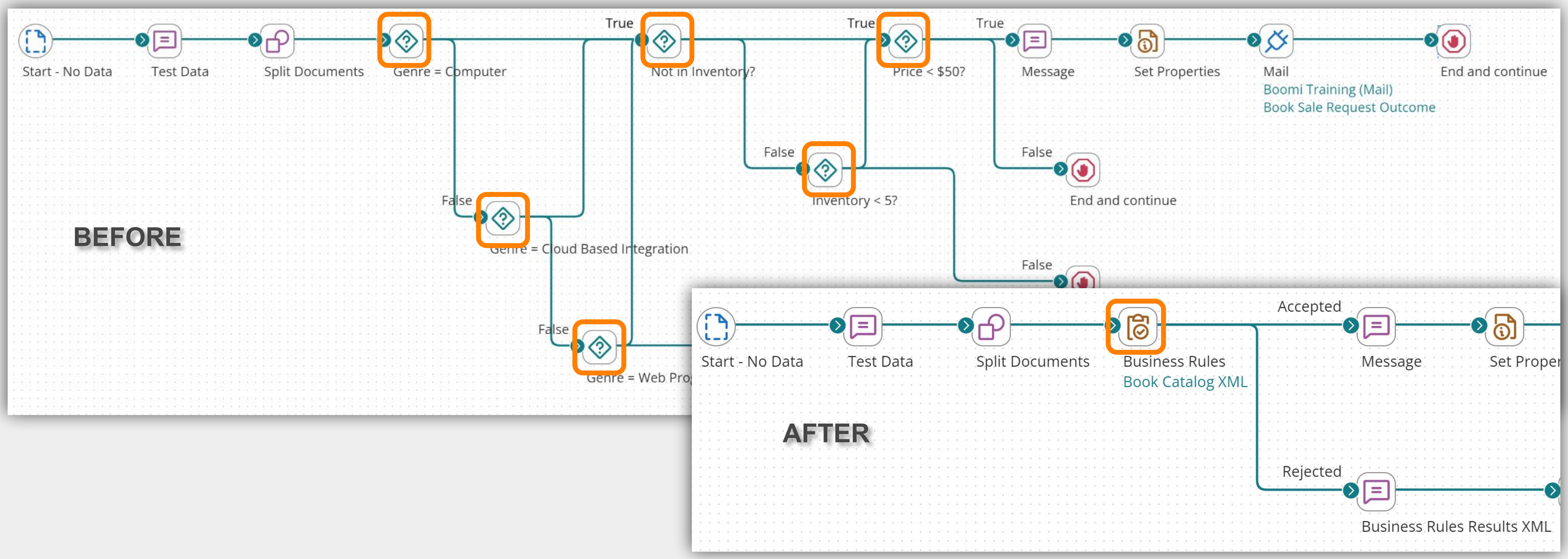
What Is the Business Rules Shape?

The Business Rules shape allows you to create advanced logic and validation rules for your process.



What Is the Business Rules Shape?

The Business Rules shape is a powerful alternative to the Decision shape because it can store multiple rules and conditions into one catch-all container



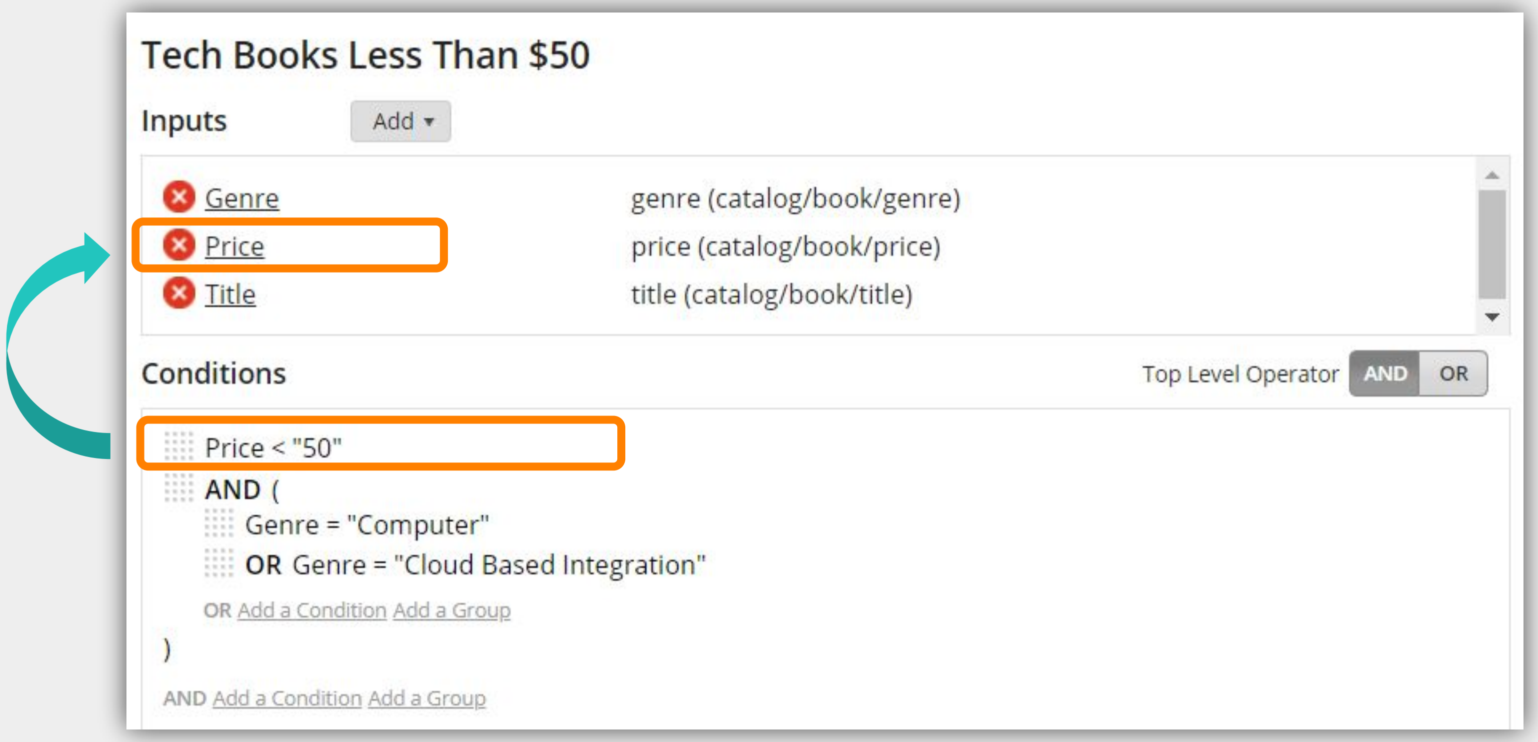
Business Rules Shape Configuration

Inputs are defined

- Field (Profile Element)
- Function

Business Rules Shape Configuration

Add conditions that reference supplied inputs



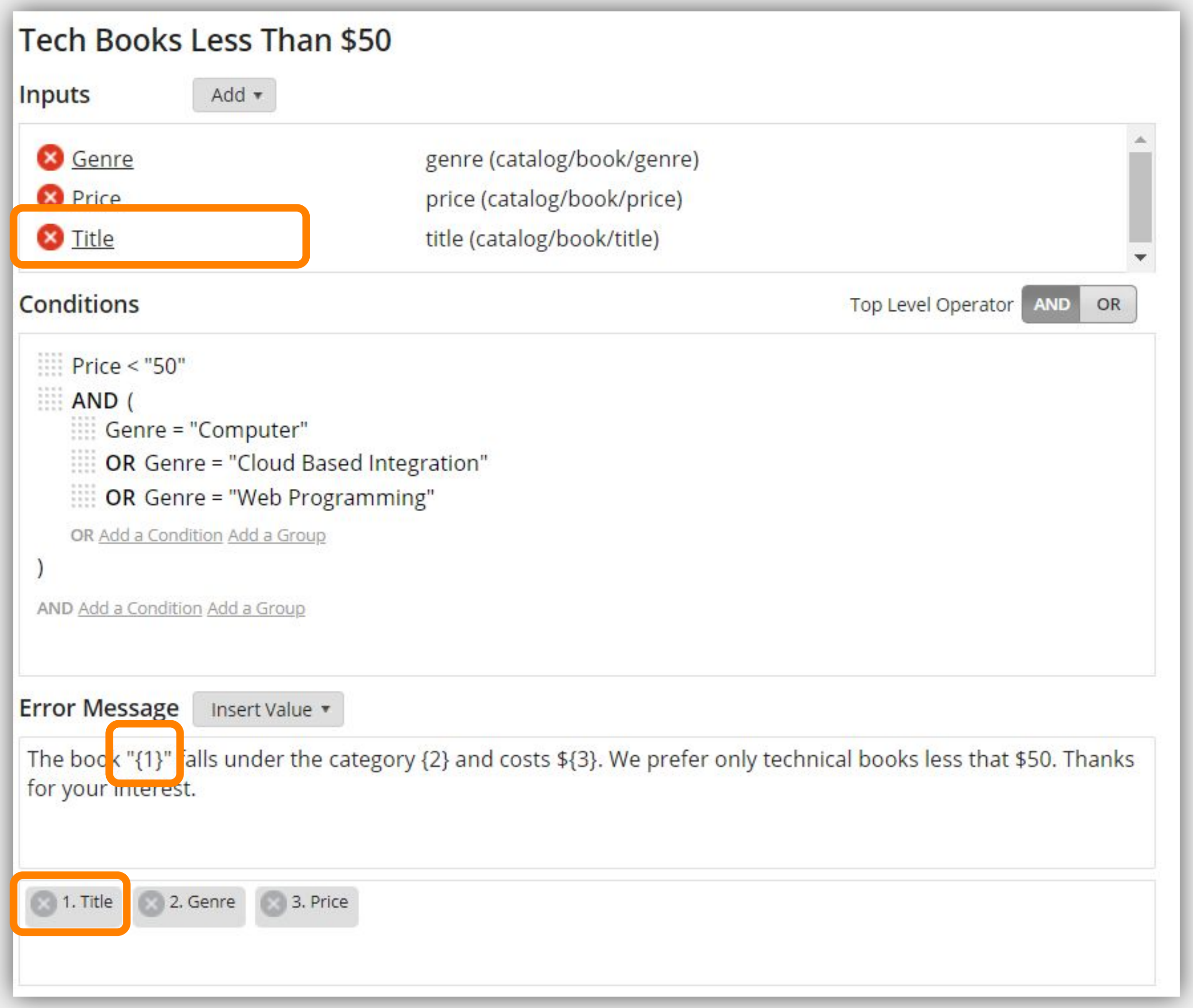
Business Rules Shape Configuration

The Business Rules shape leverages AND & OR operators and can group for compound logic implementation

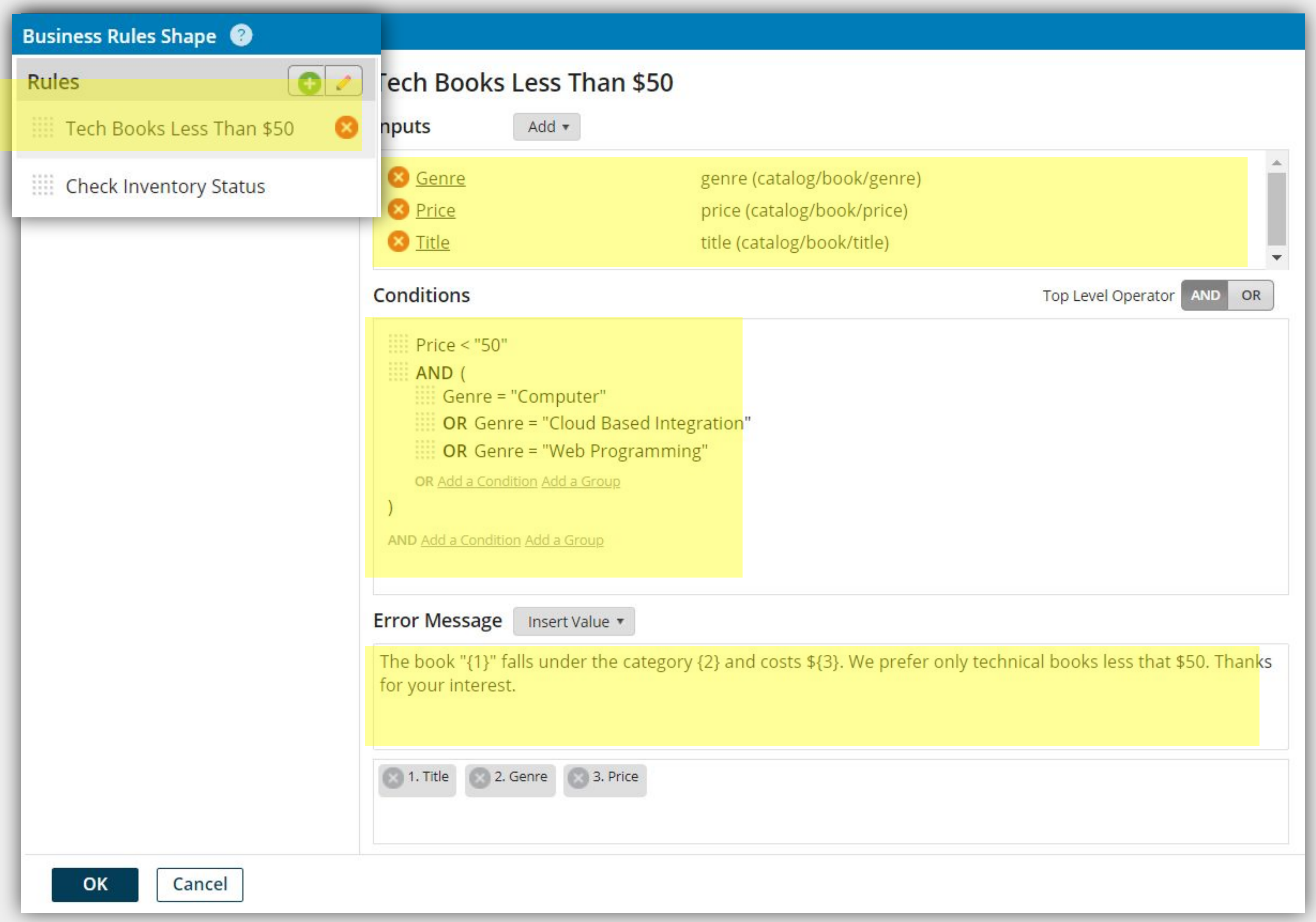
Price < ‘50’ AND Genre = ‘Computer’ OR Genre = ‘Cloud Based Integration’ OR Genre = ‘Web Programming’

Business Rules Shape Configuration

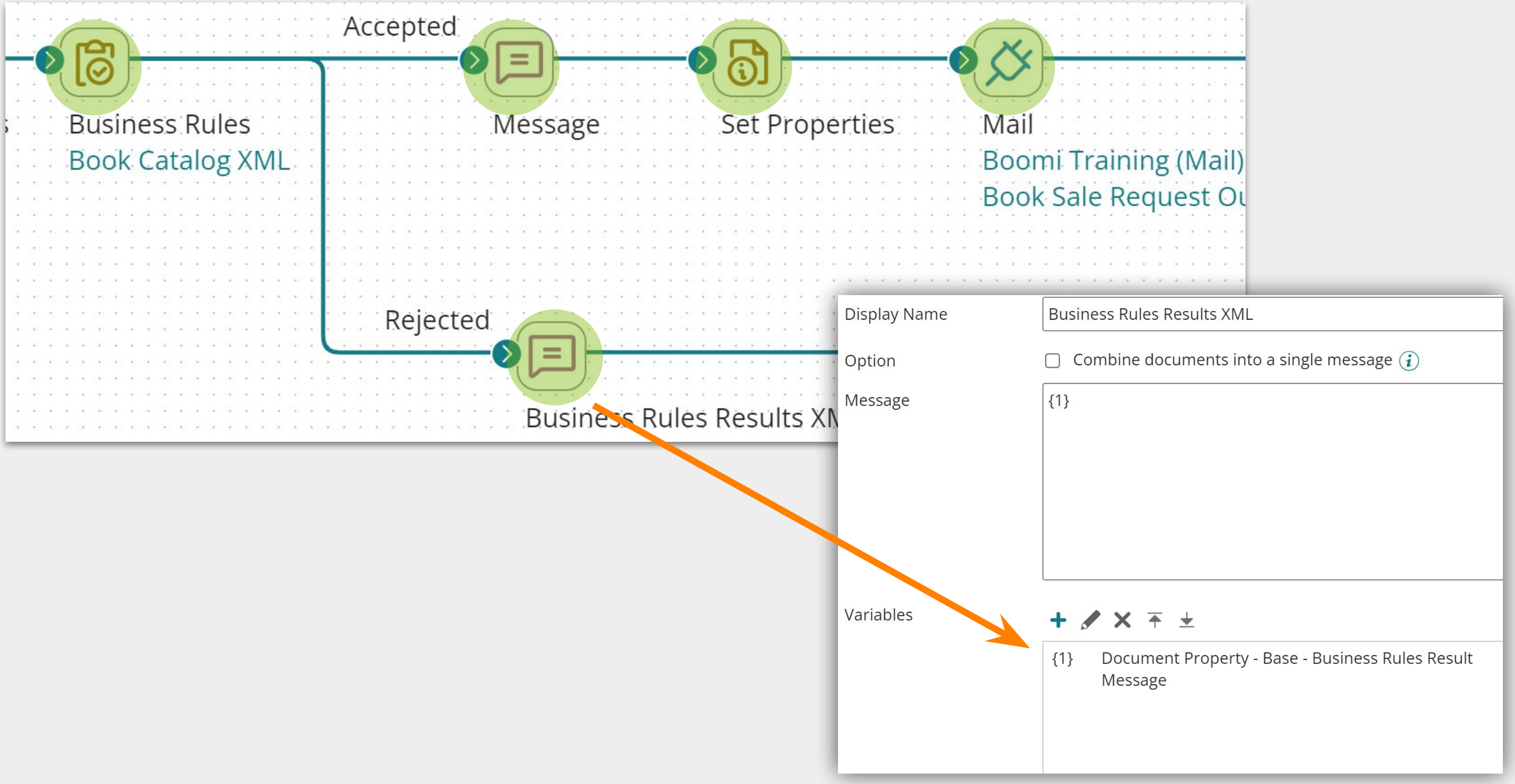
Dynamic inputs and hard-coded message data can be create a custom message.



Business Rules Shape Execution

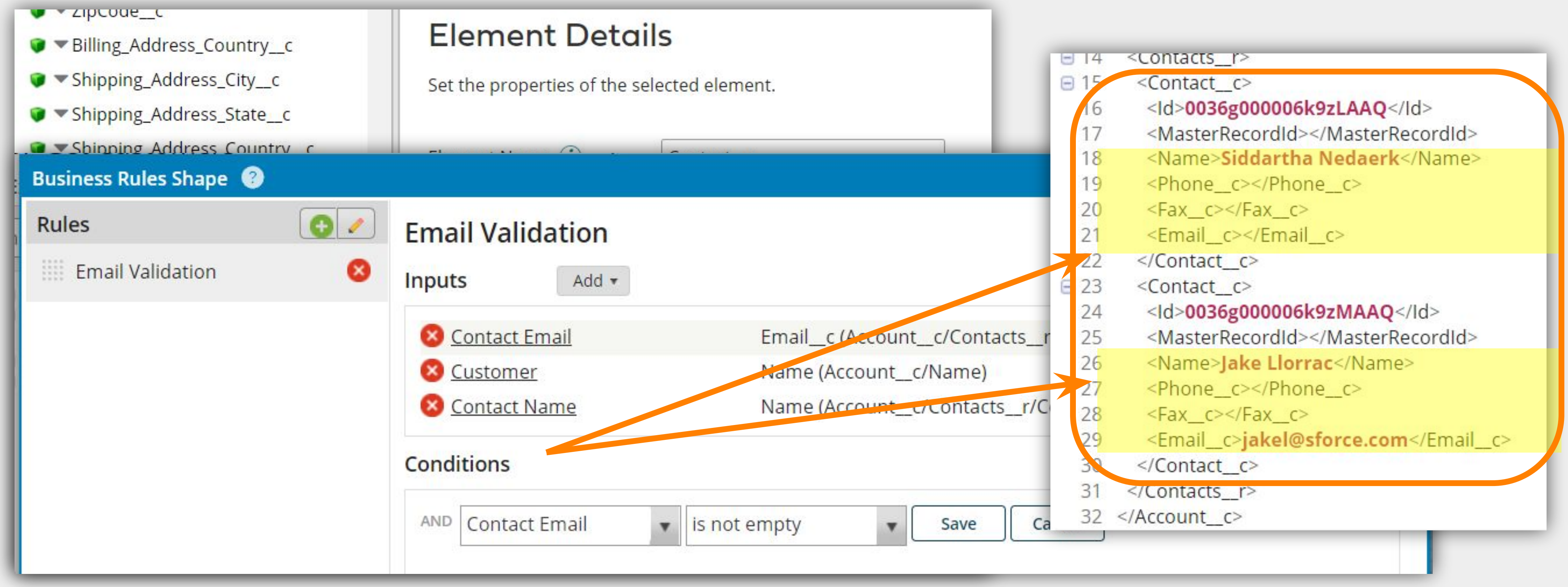


Business Rules Shape Execution



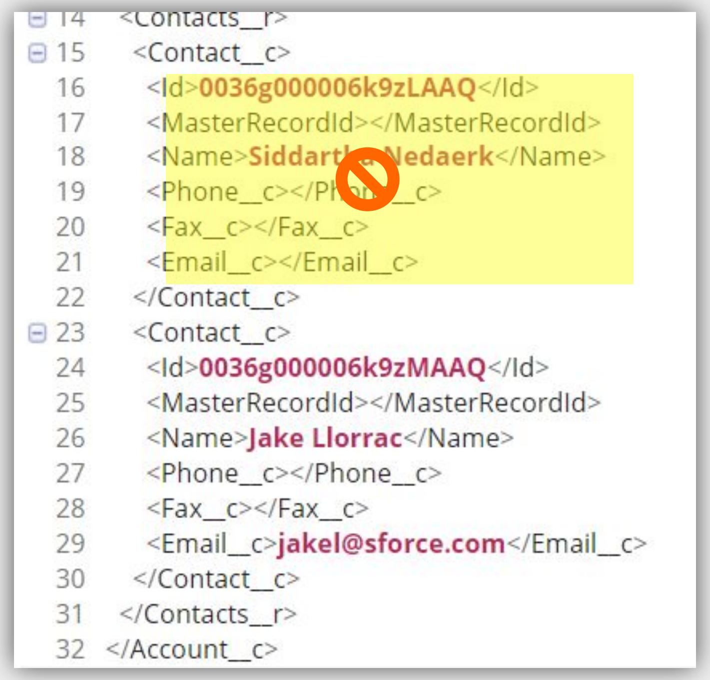
Business Rules Shape Execution

The Business Rules shape executes for the number of repeating elements used as Inputs to the rule

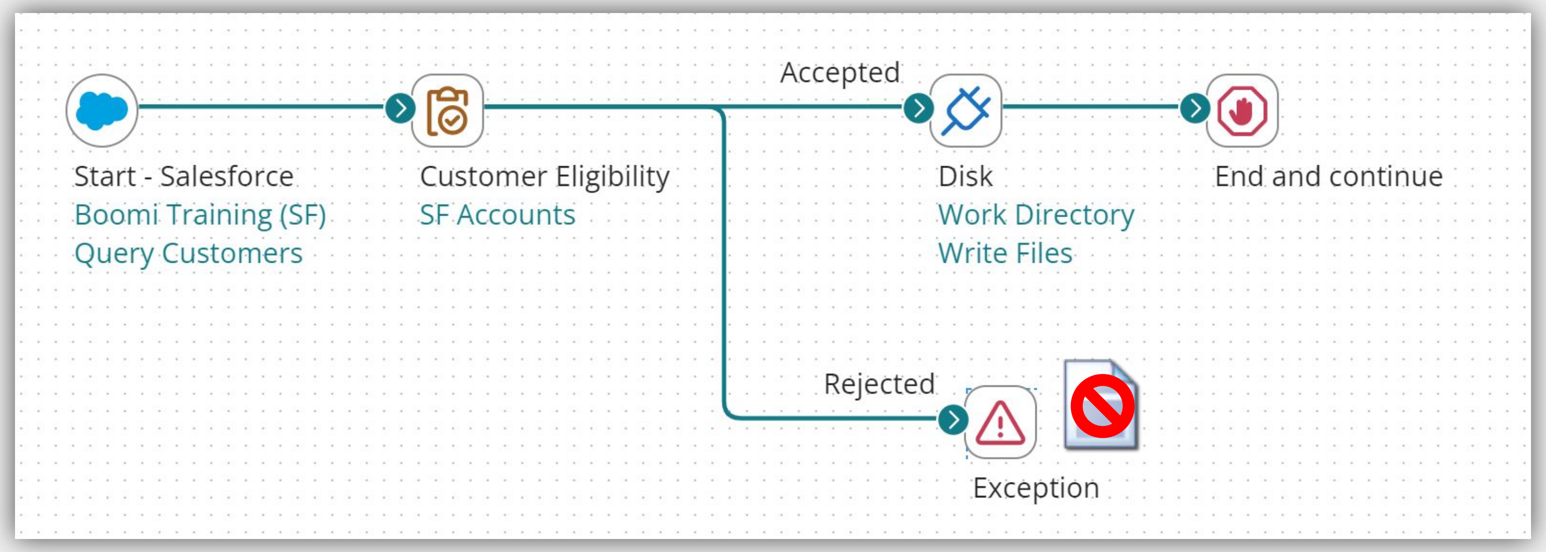


Business Rules Shape

If one repeating elements fails, it rejects the entire document.



```xml
<Contacts__r>
  <Contact__c>
    <Id>0036g000006k9zLAAQ</Id>
    <MasterRecordId></MasterRecordId>
    <Name>Siddart[illegible]Nedaerk</Name>
    <Phone__c></Ph[illegible]__c>
    <Fax__c></Fax__c>
    <Email__c></Email__c>
  </Contact__c>
  <Contact__c>
    <Id>0036g000006k9zMAAQ</Id>
    <MasterRecordId></MasterRecordId>
    <Name>Jake Llorrac</Name>
    <Phone__c></Phone__c>
    <Fax__c></Fax__c>
    <Email__c>jakel@sforce.com</Email__c>
  </Contact__c>
</Contacts__r>
</Account__c>
```



Business Rules vs Decision Shape

Business Rules	Decision
Available in Professional & Enterprise editions	Available in Standard edition
Routes documents on an accepted/rejected path	Routes documents on a true/false path
Based on the comparison of multiple inputs & conditions	Based on the comparison of two specific values
Can perform nested AND/OR logic	Single Boolean logic
Built-in error Message handling	Error handling not built in
Inputs can be profile elements and functions	Inputs can be profile elements, doc/process properties, SQL statements, stored procedures, connector calls, etc.
Executes rules for every repeating element	Only inspects first of any repeating elements
More complex to build	Simple to configure

The Business Rules shape is a powerful alternative to which of the following AtomSphere shapes?

A Flow Control

B Trading Partner

C Decision

D Cleanse

Conditions within a Business Rule input values. These Inputs can derive from all of the following except:

A Profile Fields

B Document Logs

C Static Values

D Functions

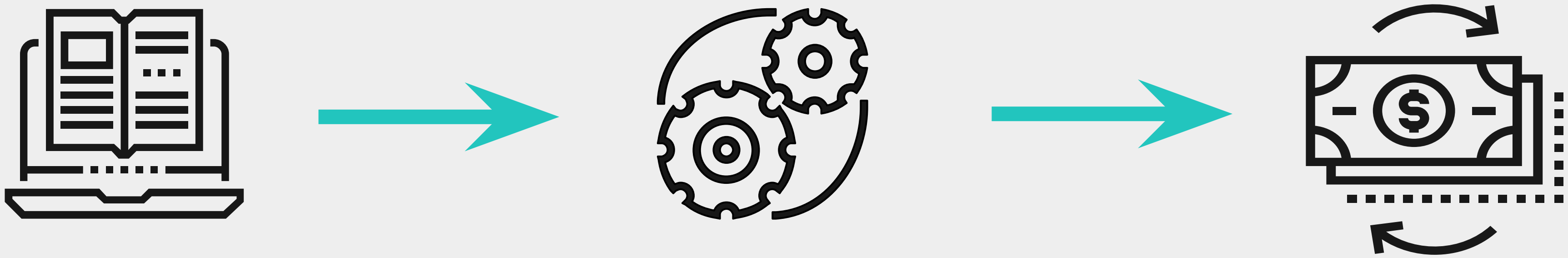
Assume a Business Rule's condition uses a Profile input consisting of a repeating element. If one of those elements fails the Business Rule then the entire Document is rejected.

A. True

B. False

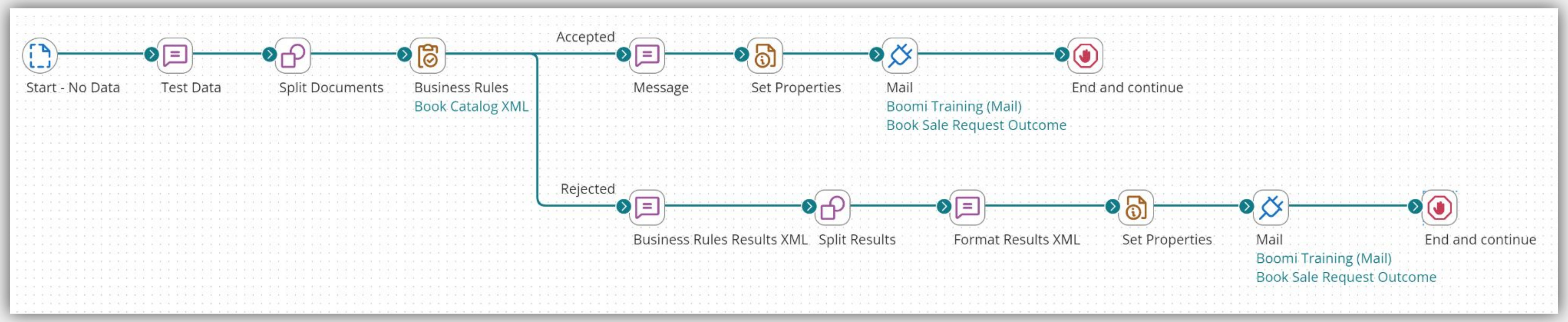
Business Use Case for Activity

- Athens Books buys used books from other booksellers
- Only specific prices and genres are accepted if inventory is low.



Business Rules Activity

- Replace the Decision shapes with one Business Rules shape
- Configure a Business Rule with Inputs, Conditions, and Detailed Error Messages
- Add a Data Process shape to parse the Business Rules Results XML
- Configure the Message shape to handle error messaging
- Send an email message with results



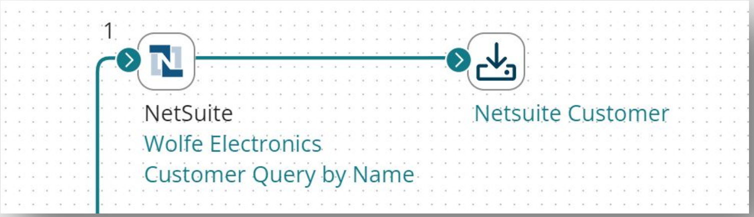
HANDS ON ACTIVITY

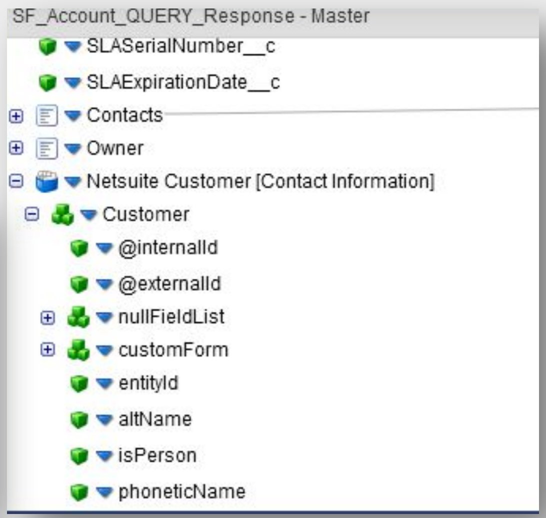
Business Rules Activity

Document Cache

What is Document Caching?

- Add documents to a cache to reference later in a process or subprocess execution.
- You can index documents in a cache, which are available for the entire process execution
- Enables multi-source profile mapping using cached documents





Document Caching

- Add to Cache shape
  - Add Documents to a Document Cache to reference later in a processes
- Document Cache component
  - Reusable component for index and key definition
- Add Cached Data – mapping feature
  - Allows you to include multiple source documents in a single map

Document Caching (Continued)

- Retrieve from Cache shape
  - Used to get documents from a document cache

- Remove from Cache shape
  - Used to remove documents from a document cache

- Document Cache lookup
  - Performs a dynamic lookup against a Document cache component to retrieve particular fields from a document

When using the Add to Cache, you are given the option to persist profile data for subsequent Process executions.

A True

B False

The ‘Add Cached Data’ feature is found in which of the following shapes:

A Set Properties

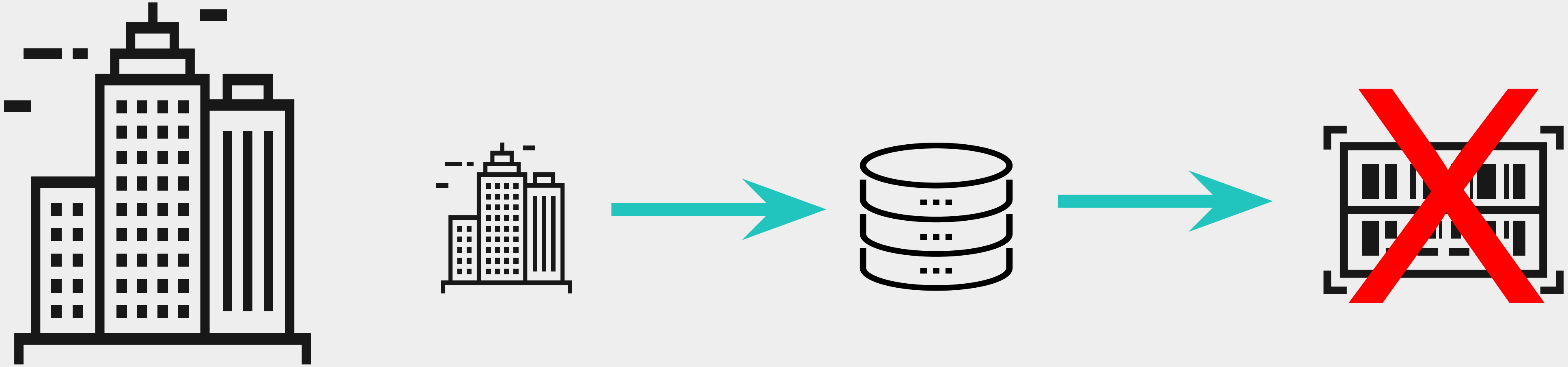
B Map

C Data Process

D Load from Cache

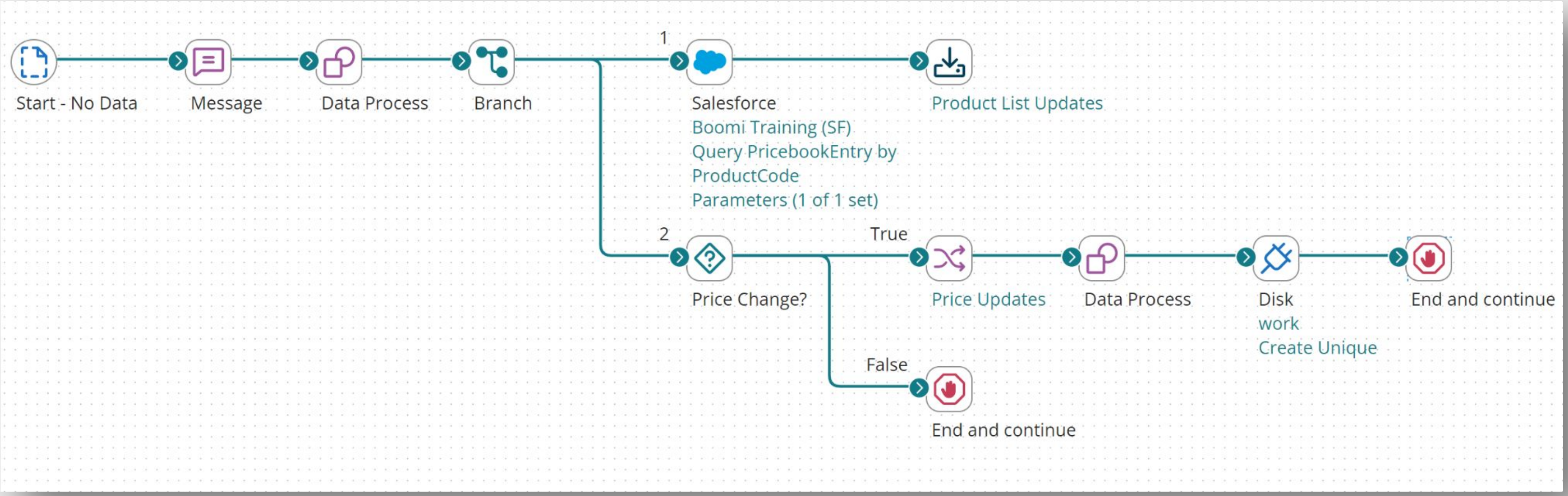
Business Use Case for Activity

- Canyon Enterprises (CE) stores their product information in Salesforce
- Campus Store, an independent reseller, carries many of Canyon's products but keeps their own pricing database in-house
- Campus's prices can get out of sync with Canyon's



Document Cache Activity

- Process invoices containing product codes and prices
- Cache the current product codes and prices from Salesforce
- Match the product codes, then compare in-house prices to Salesforce
- Update thein-house price list if it differs from Salesforce
- Write the updated file to disk



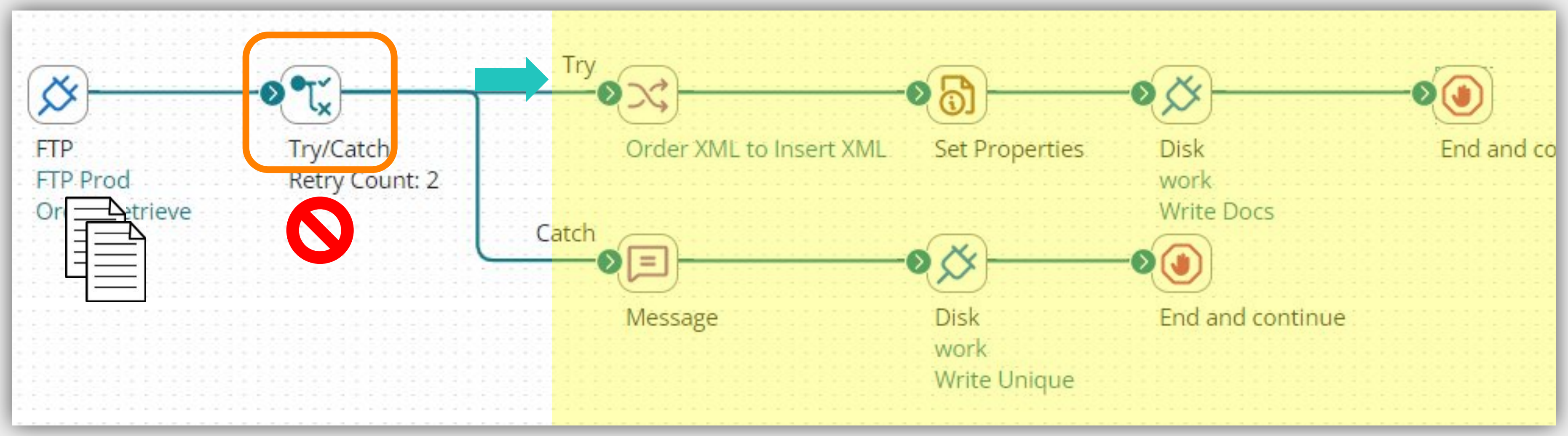
HANDS ON ACTIVITY

Document Cache Activity


Try/Catch

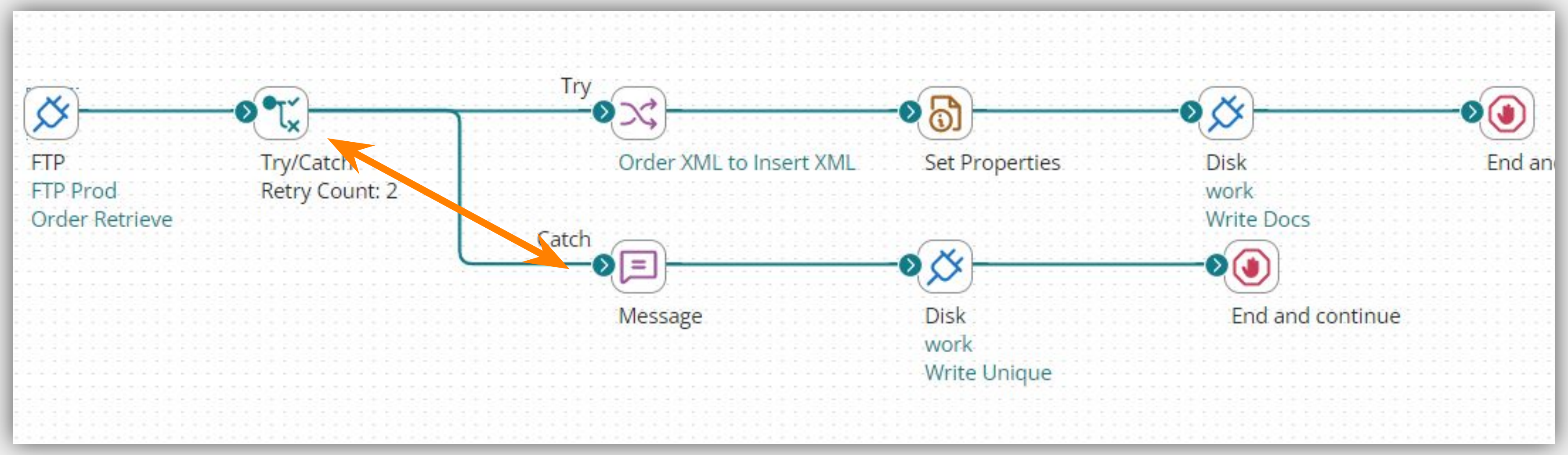
What is the Try/Catch Shape?

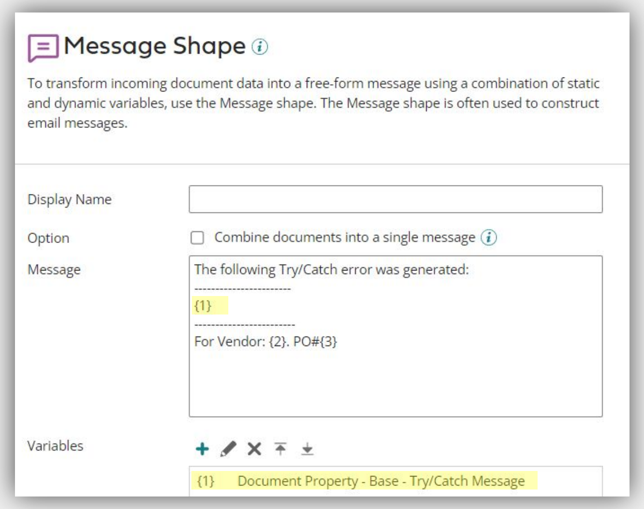
- The Try/Catch shape captures document-level errors for one or more documents which fail during execution
- It must be placed BEFORE the main processing shapes in your process where you hope to capture the error
- All documents flow down the **Try** path to be processed. Failed documents are caught and sent down the **Catch** path



What is the Try/Catch Shape?

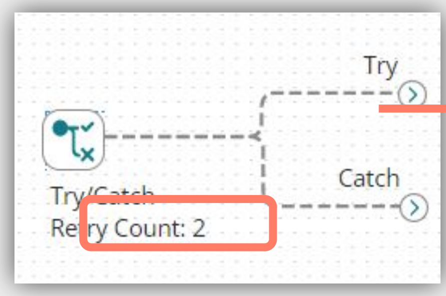
- Designs advanced logging and processing for failed documents.
- Prevents process-level failures in the event a document fails, allowing you to act upon it later in your process flow.
- Captures the exception using the Message or Notify shape and include a parameter to refer to the Try/Catch message document property.

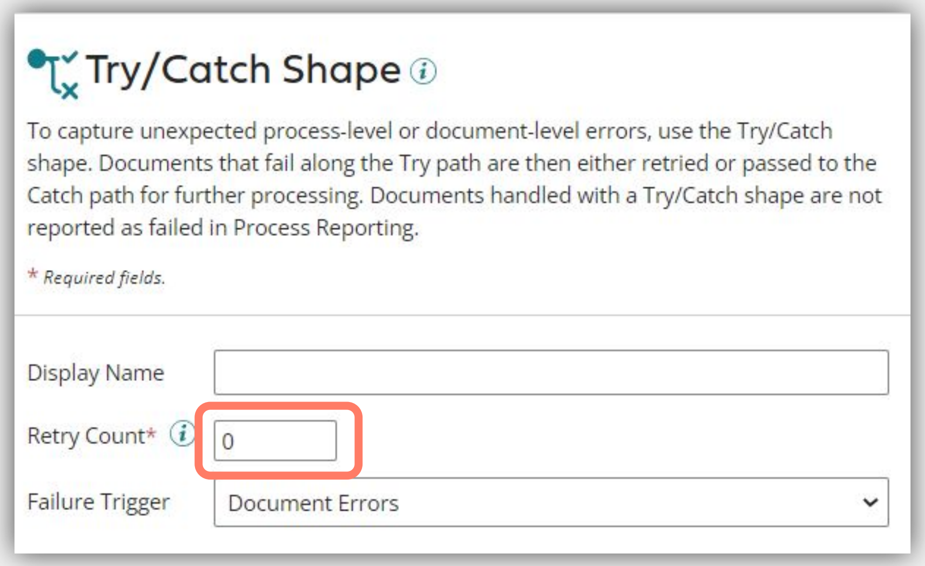




Try/Catch Shape - Retry

- You can retry failed documents.
- The Retry Count defines the maximum number of times a failed document is retried through the main series of shapes AFTER the Try/Catch shape
- Can be set from 0 to 5
- The Retry Count option pauses between retries



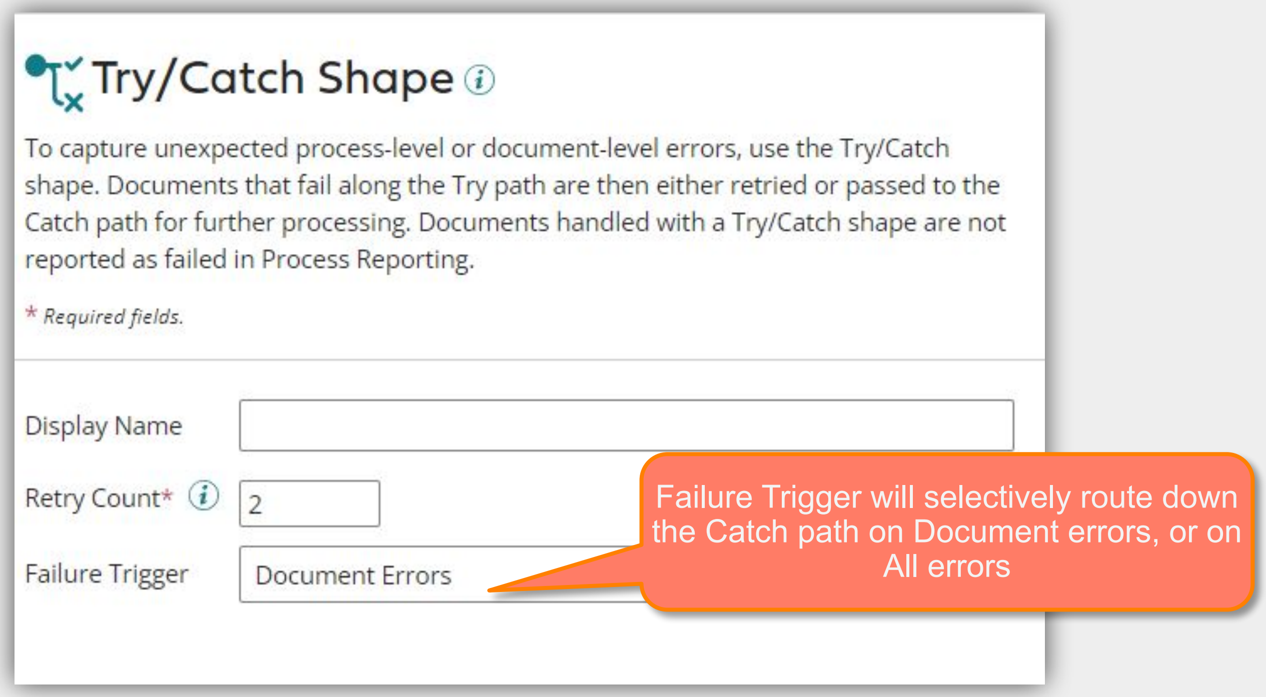


Try/Catch Shape – Retry Schedule

Retries	Timing
0	The document is not retried. Try/Catch uses it to catch the error and deal with it in another way, i.e., archive the inbound document, send an email alert, etc..
1	The first retry occurs immediately following the first document failure.
2	If the document fails a second time, the system waits 10 seconds before retrying.
3	If the document fails a third time, the system waits 30 seconds before retrying.
4	If the document fails a fourth time, the system waits 60 seconds before retrying.
5	If the document fails a fifth time, the system waits 120 seconds before retrying one last time.

Try/Catch Failure Trigger: All Errors

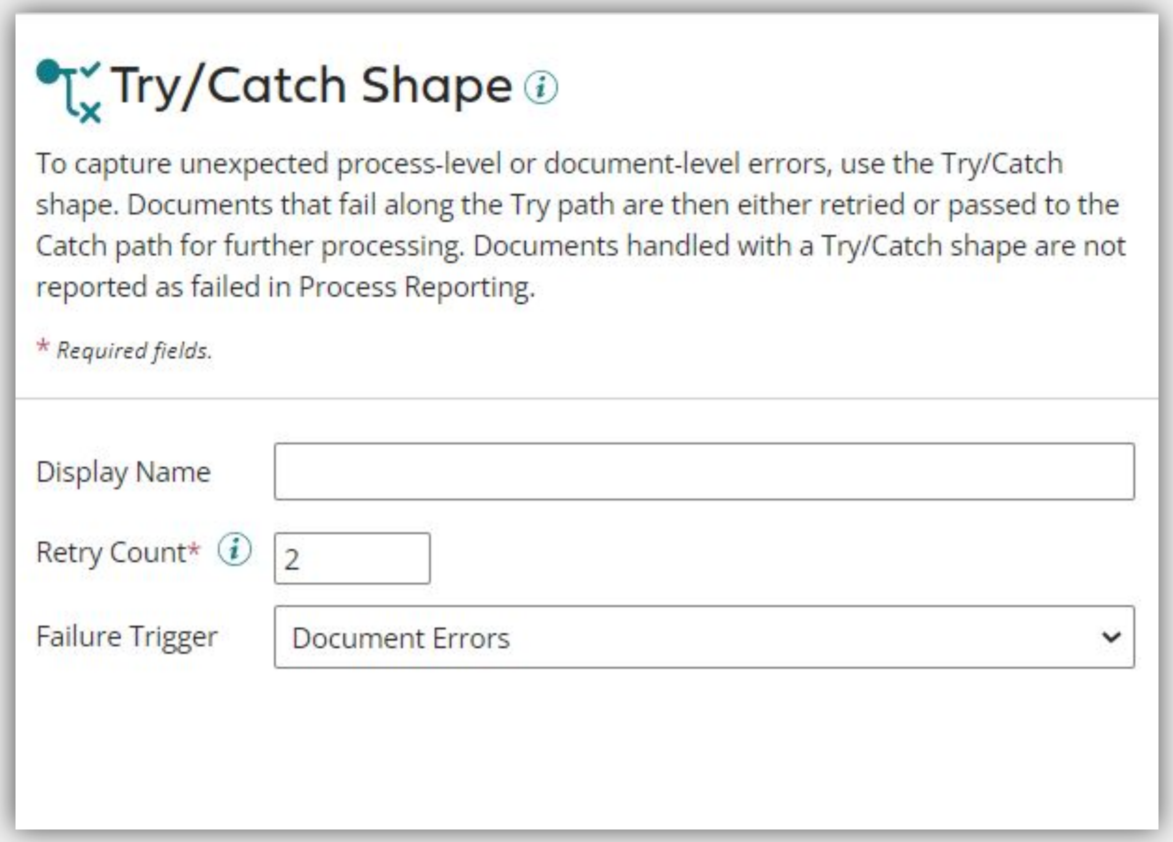
- Try/Catch shape can be configured to catch just Document Errors or All Errors.
- The following errors are captured, no matter how you set the Failure Trigger field:
  - A connectors logging errors for specific documents when errors occur, e.g., if an update operation fails because the document has an invalid ID.
  - An Exception shape with the Stop Single Document checked.
  - A Map shape, if it does not produce any output, generates a validation or connector error.



Try/Catch Failure Trigger: Document Errors

The following errors are ***not*** captured if the Failure Trigger field is set to Document Errors:

- NullPointerExceptions and similar process-level errors
- An Exception shape that has the Stop Single Document check box turned off
- Connectors return a full exception error to the process instead of capturing the error internally and logging it per document



Try/Catch: The Catch Path

Consider the following actions in the Catch path:

- Capture the source record for reprocessing
- Mark the source record as in–error
- Notify a user – targeted notification
- Format a message using the Try/Catch error document base property
- Save or archive the failing document
- Use an Exception to cause an error event, once other actions have been taken
- Decision or Route based on type/severity
- Call a standard error handling sub-process

In order to catch Documents which error at the Map shape, the Try/Catch shape should be placed:

A. Within the Map

B. Before the Map

C. After the Map

D. Anywhere in the Process

Assume that 5 Documents are being processed through the Try/Catch shape. If 2 of the 5 Documents fail, then which of the following happen? (Choose all that apply.)

- **A** The 2 failed Documents will complete the Try path
- **B** The 2 failed Documents will complete the Catch path
- **C** The 3 successful Documents will complete the Try path
- **D** The 3 successful Documents will complete the Catch path
- **E** None of the Documents are processed

Within the Try/Catch Retry Count option, what is the maximum number of retries that can be configured for a failed Document?

A | 1

B | 2

C | 5

D | 10

E | 15

Which of the following is NOT an advantage to using the Try/Catch shape?

A Allows you to design advanced logging and processing for failed Documents.

B Prevents process-level failures in the event that a Document fails so that you can act upon it later in your Process flow.

C Allows the Process to complete more quickly.

Which shape is often paired with the Try/Catch shape in order to convert the Document into a free-flow text Document containing details about the Try/Catch error message?

A Set Properties

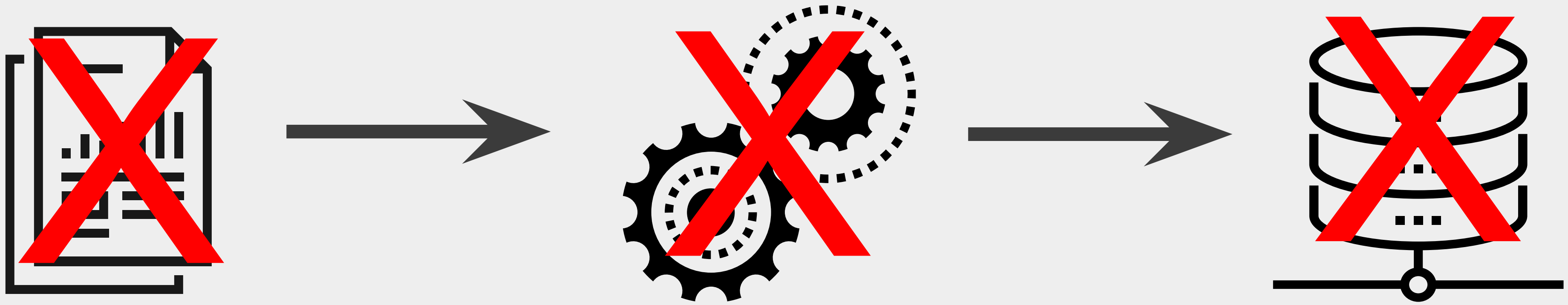
B Map

C Notify

D Message

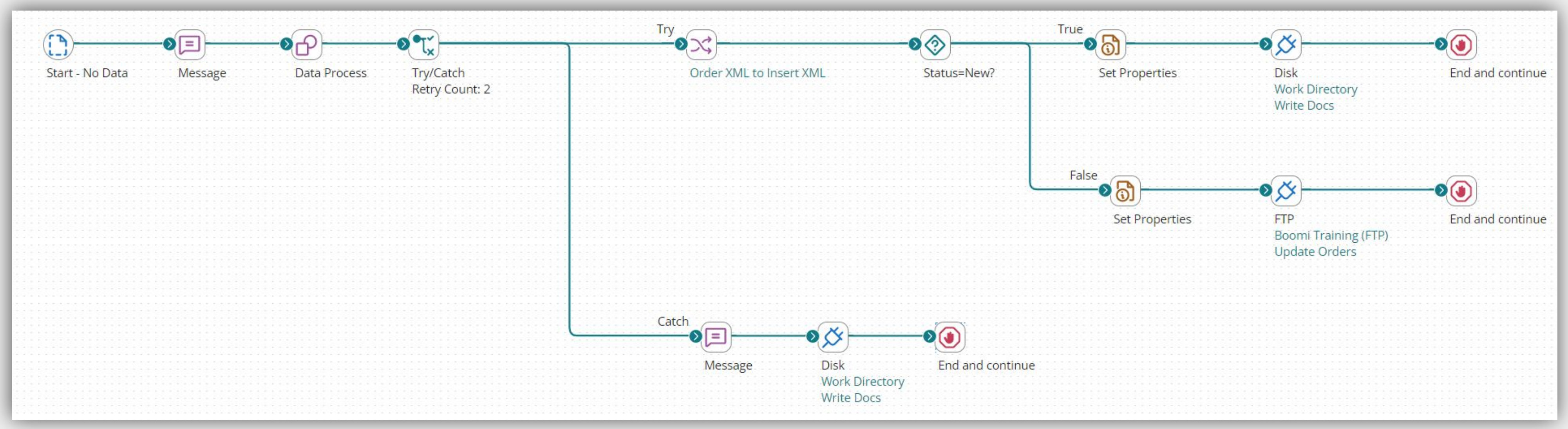
Business Use Case for Activity

- Purchase Orders (PO) are not being processed in AtomSphere
- A batch with three PO’s was halted, and it wrote none of the PO’s to the database.



Try/Catch Activity

- Execute the process in Test mode to see errors
- Add a Try/Catch shape to catch documents containing errors
- Configure a Message shape to capture detailed errors
- Write detailed error logs to disk for archiving
- Route successful documents down the try path



HANDS ON ACTIVITY

Try/Catch Activity

Error Handling

Error Types

- Process Errors
- Document Errors
- Atom Runtime Errors

- Recoverable vs. Unrecoverable Errors
    - Most AtomSphere errors are recoverable with good process design
    - Some AtomSphere errors, such as a server crash, are unrecoverable
    - Unrecoverable errors halt process execution

Process Error Impact

Halts process and all documents immediately

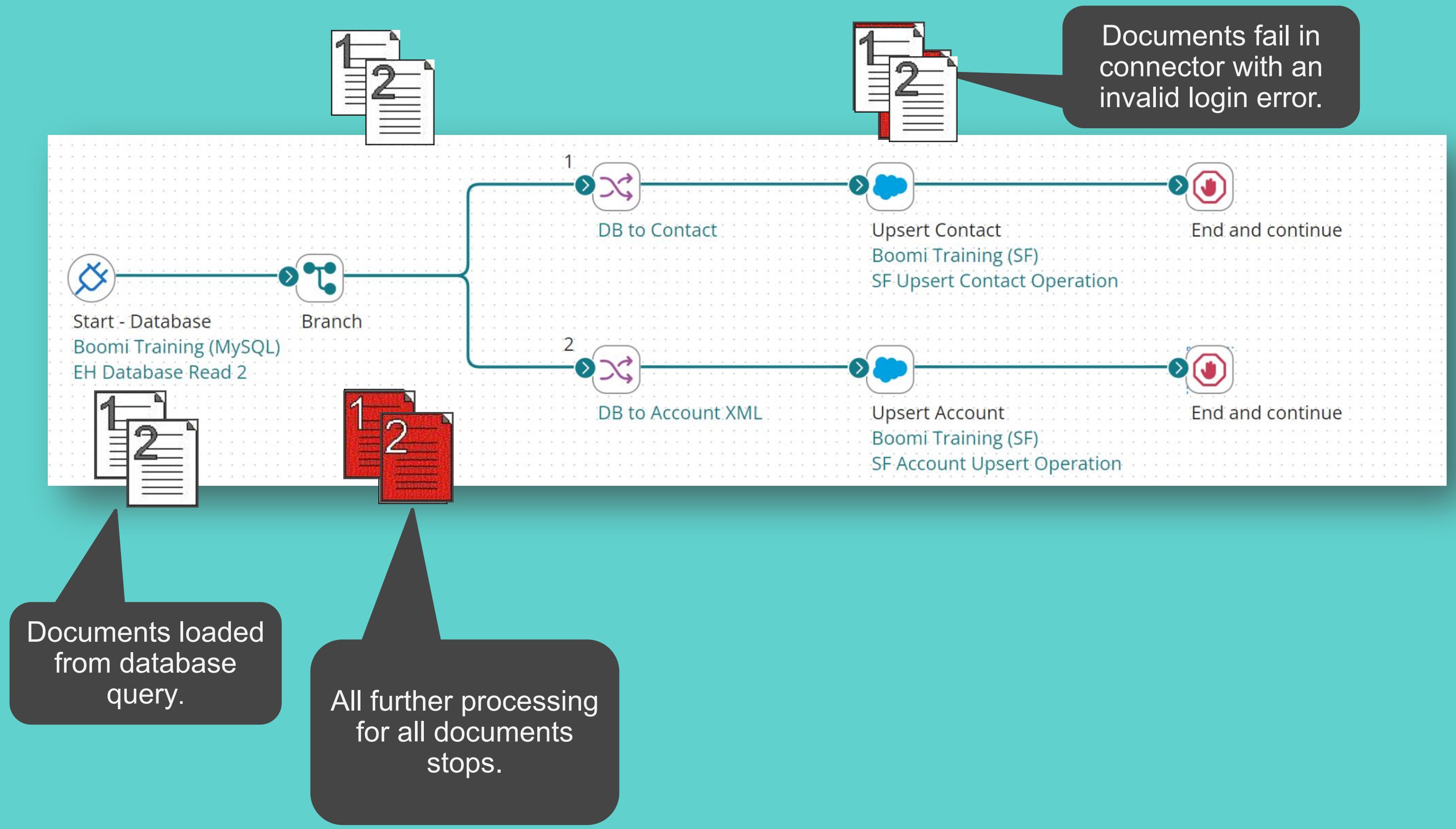
No checkpoint or commit mechanism = no roll back

No further processing, unless Try/Catch is used

Process Error Causes

- Connector connectivity errors (varies by connector)
- Connector app responses (varies by connector)
- Exception shape – when “Stop a Single Document” is unchecked
- Unhandled Data Process scripting exceptions
- Incomplete process/component configuration, e.g., NullPointer (dev-time fix)
- Connector errors on the Start shape (cannot be caught, because there are no documents)

Process Error Example



Document Error Impact

Will halt the document or documents with the error

No further processing, except if Try/Catch is used

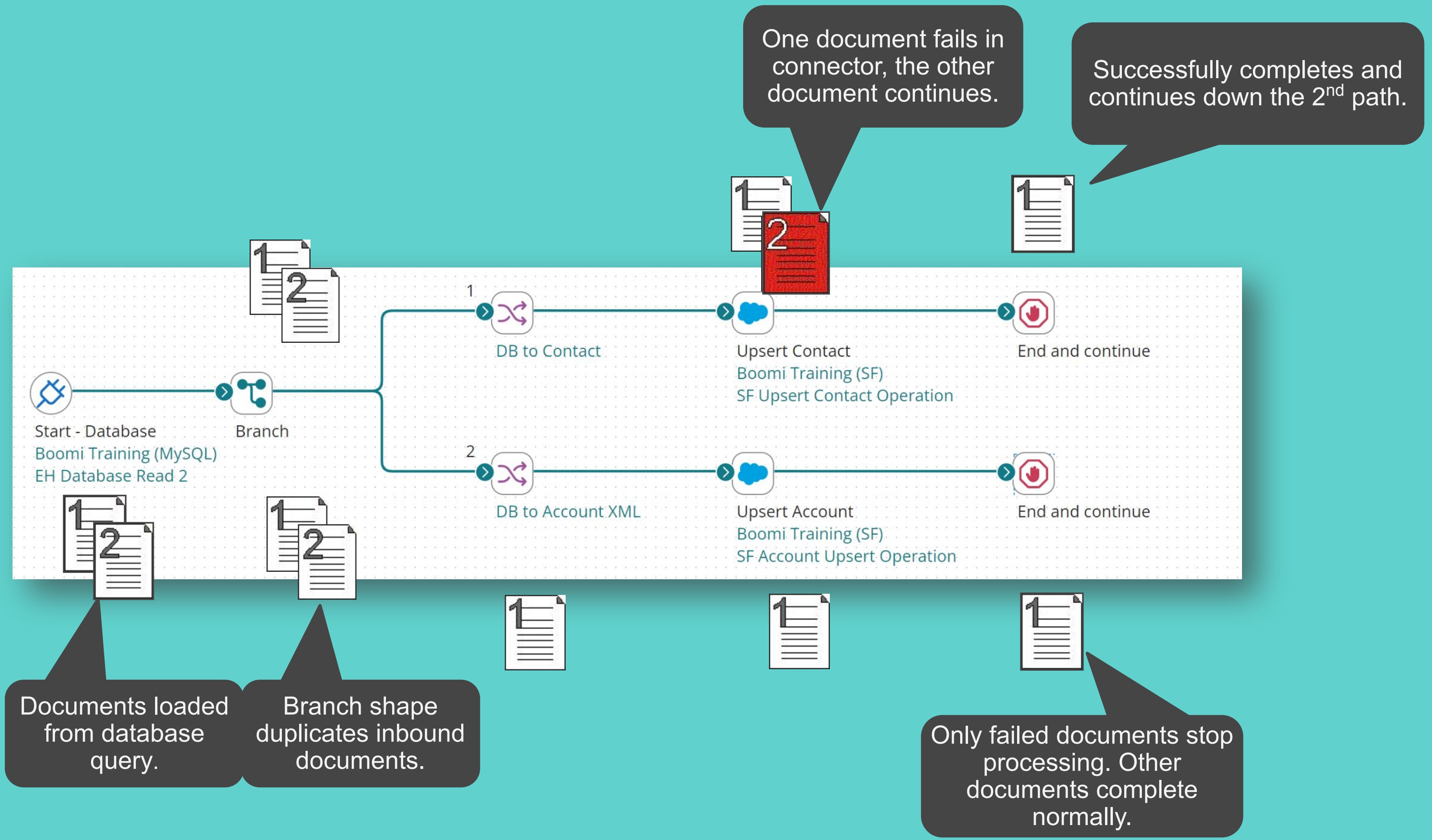
Successful documents continuing processing

No rollback for any previously completed shapes

Document Error Causes

- Mapping/data errors (function failures, profile mismatch/no output, data formatting/missing required fields), script errors, data format errors.
- Connector connectivity errors (varies by connector).
- Connector application responses (varies by connector, but more common at document level).
- Exception shape - when checking Stop a Single Document.

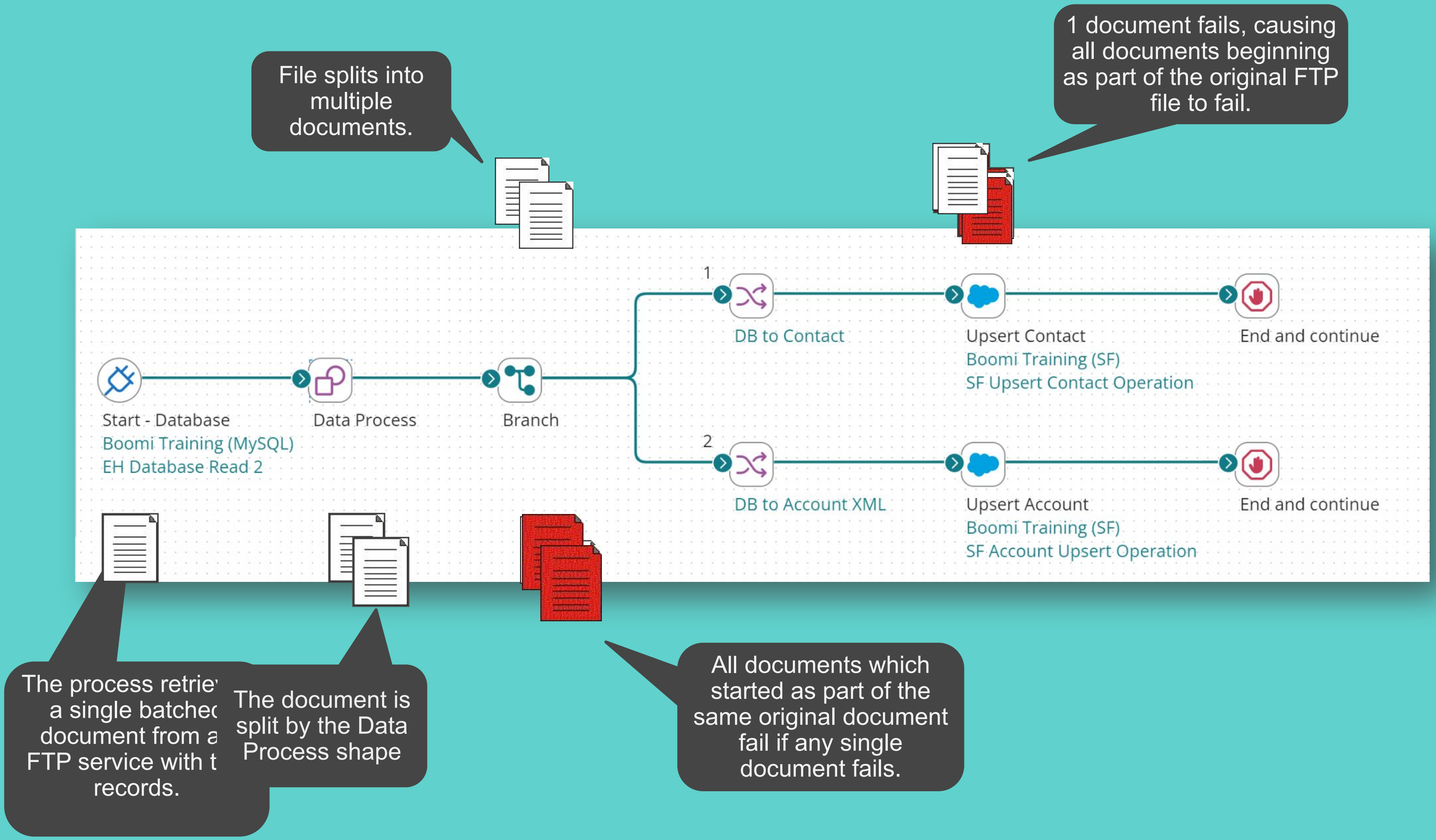
Document Error Example



Errors Impact Document Flow

- When original document fails, this impacts:
  - All logical records contained in the original document
  - Documents trace back to the source document
- Batching and splits can impact the scope of document error failure
- Try/Catch changes the impact of document errors by resetting the document IDs so only failed document stops processing

Document with Split Error Example



Error Type: Atom Errors

Stops the Process

- Not Recoverable
- Cannot be caught

Handled using runtime monitoring

Causes a Process-Level Error

- Document processing is stopped
- Failure reported when Atom online

Error Type: Atom Errors

During Process Execution

- Activities at Atom level fail
- Considered an Atom error
- Caused by Lack of Resources

Examples

- JVM running out of heap space
- Atom runs out of memory or writable disk space
- The server is disconnected

INSTRUCTOR TO DEMONSTRATE

Error Handling - Activity 1

Common Error Handling Situations

Validation Techniques

Avoiding Errors: Validation Techniques

Decision

Routes documents using a true/false comparison of two values

Business Rules

Works with the profile structure of a document

Allows for multiple rules to accept or reject a document.

Cleanse

Enables document field value evaluation to determine whether to repair or reject the document before further processing

Validation Technique Comparison

	Multiple Tests	Customized Error Message	Two Branches	Single Test	Simple Syntax	Repair or Reject
Business Rule	✓	✓	✓			
Decision			✓	✓	✓	
Cleanse	✓		✓			✓

A Try/Catch resets the Document ID for all documents that flow down the Try path.

A True

B False

This shape has a built-in Error Message function which allows you to create a customized error message and assign the message to the Base document property.

- A. Business Rules
- B. Decision
- C. Exception
- D. Cleanse

Error Handling Techniques

Technique: Exception Shape

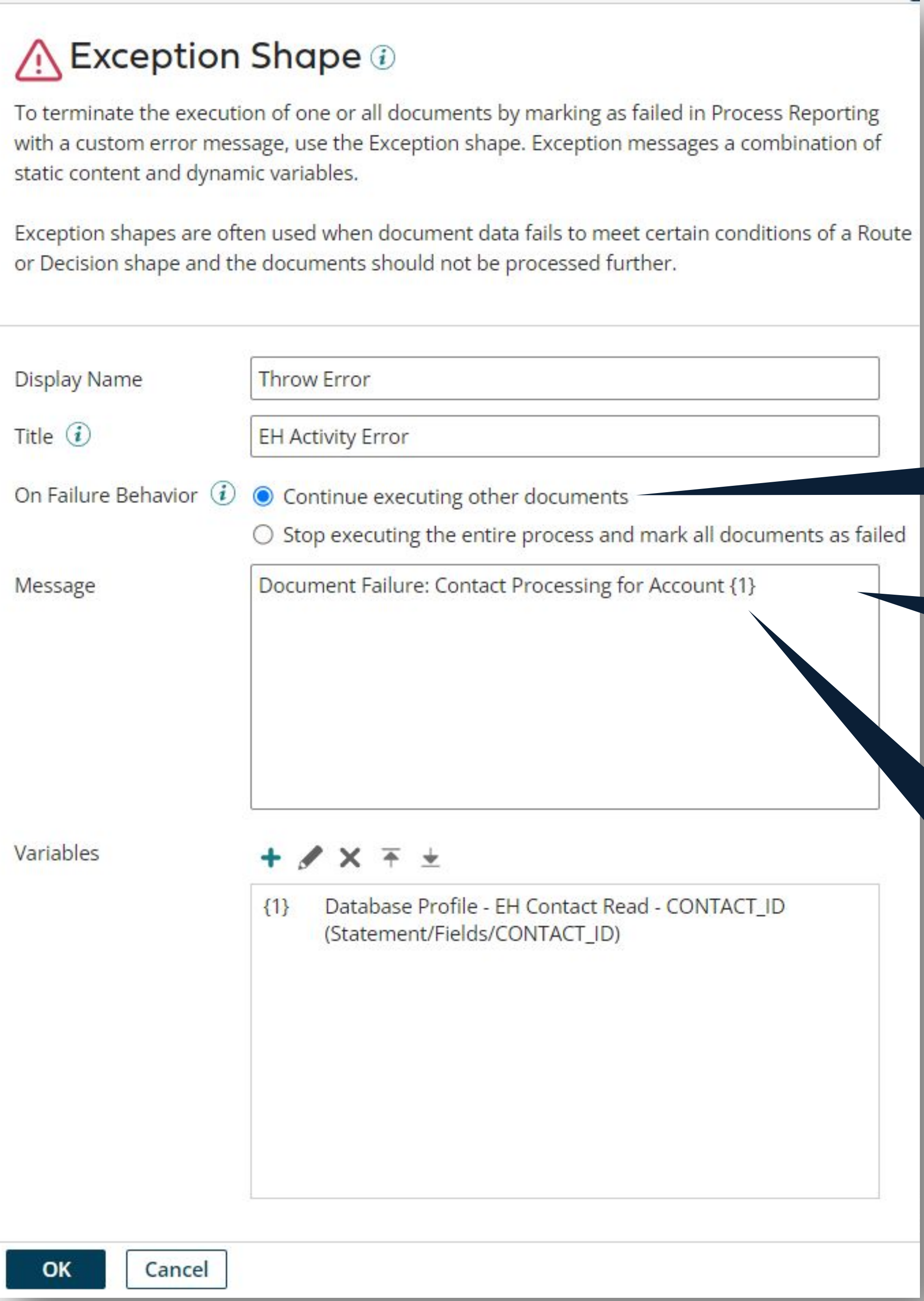
Exception shape

- Terminate the data flow down a path
- Define custom error messages to be reported on the Process Reporting page
- It is often used with the Route or Decision shape when a document does not meet specific criteria.

Custom error messages

- Can be a mix of static and dynamic content
- Dynamic content is populated using parameters
- Parameters can represent values such as:
  - Data from a document field
  - Current system date/time
  - Static values
  - Results of a database query
- Can use multiple parameters when creating a message

Technique: Exception Shape



Checked, stops only an individual document

It returns the message to logs and events.

Parameters are inserted using numbers and brackets in Message

Technique: Notify Shape

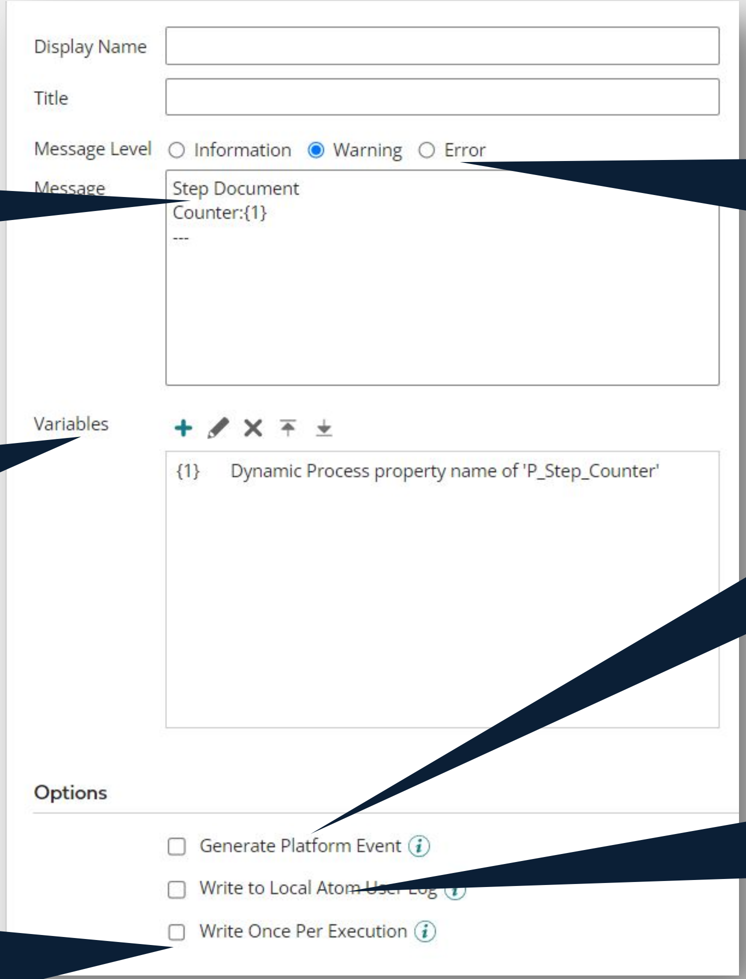
- The Notify shape enables you to build custom execution logs and/or send customized messages to your subscribed email alerts or RSS feed
- Notifications can be static messages or include dynamic content from input parameters
- Creates one or more notifications at the document level for each process execution
- Data is aggregated into a single message, then sent to the user’s email or RSS feed after process completion
- The Notify shape:
  - Used inline within a process
  - Does not alter your data
  - Designed to read any relevant document fields or properties, then pass the document along to the next shape untouched

Technique: Notify Shape

A custom message containing static and dynamic content.

Variables provide one or more values to insert into placeholders defined in the notification

If selected, individual log entries or event messages are generated per document that reaches the Notify shape are aggregated and written only once.



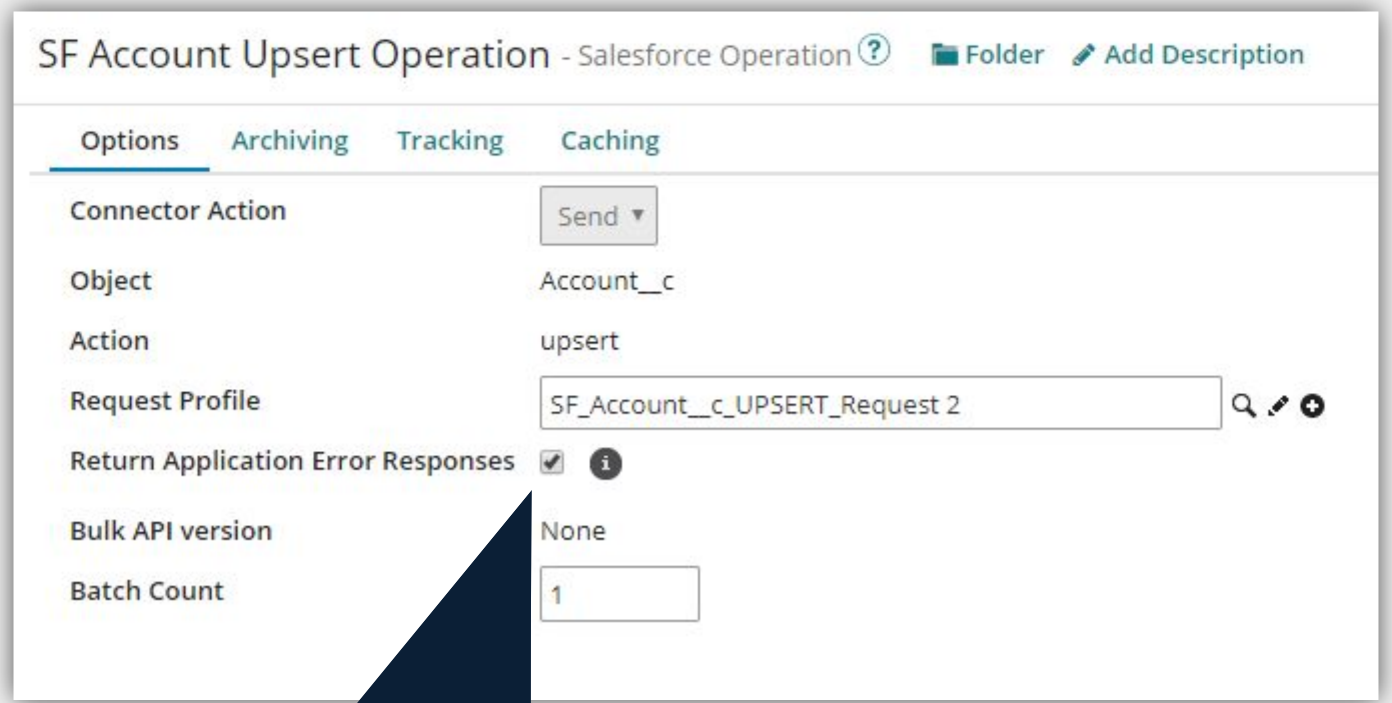
The type of notification being generated: Information, Warning or Error.

If selected, the notification is sent as an email alert to all subscribers.

If selected, it writes the notification to the base log folder of the local atom.

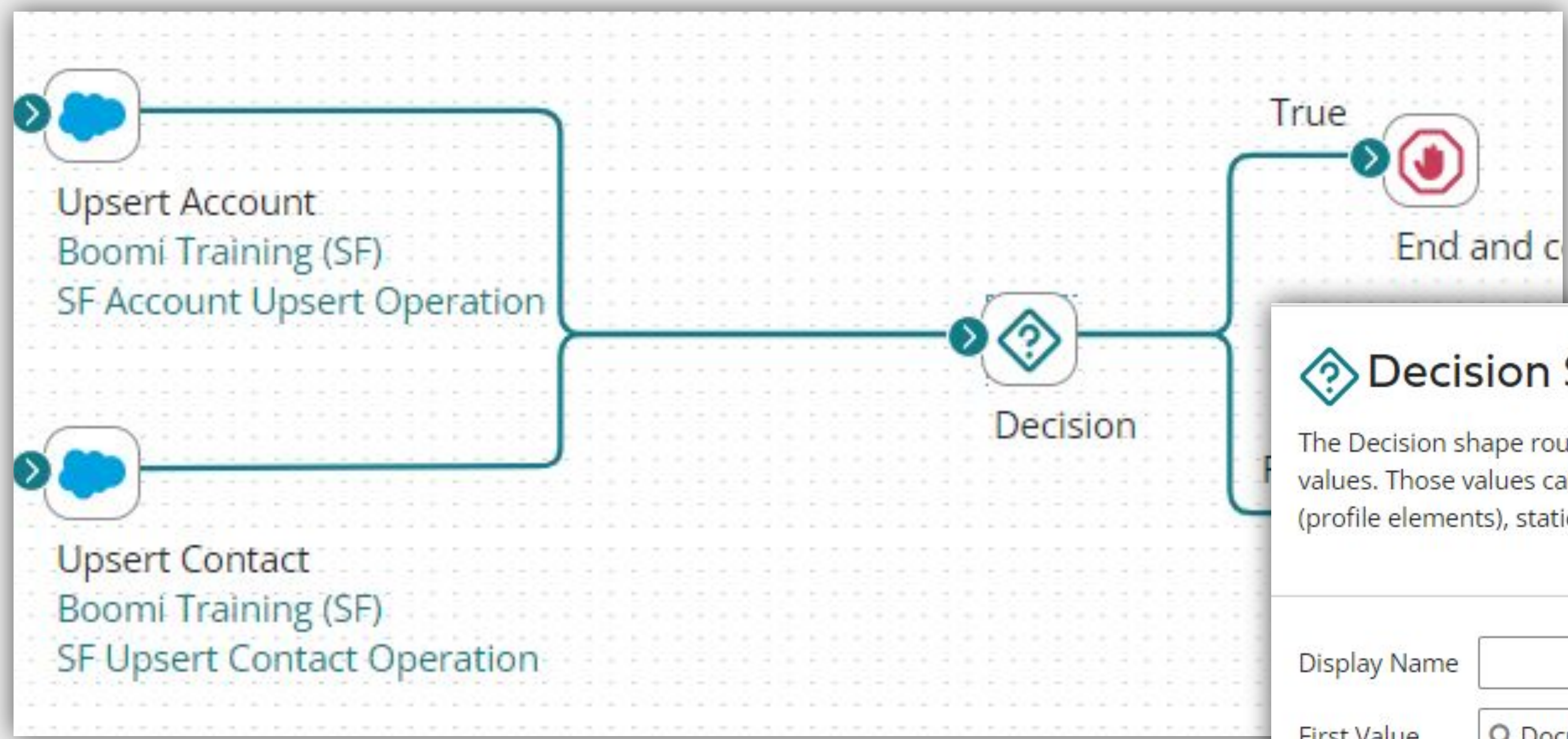
Technique:
Connector – Application Error Response

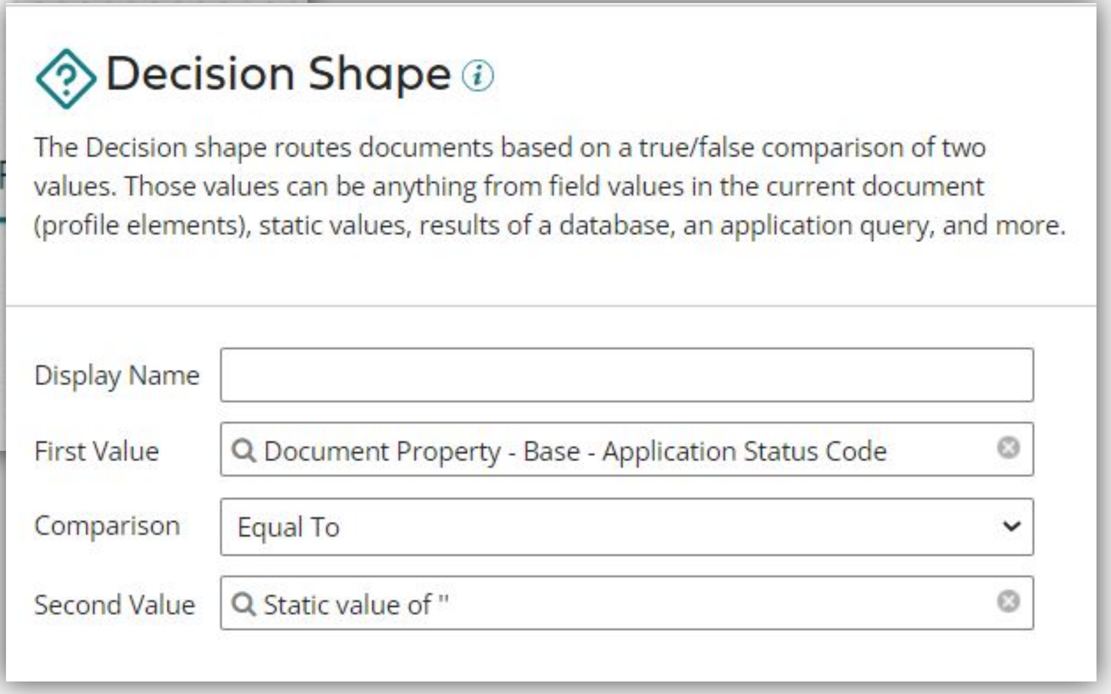
- Some connectors will return an error response to handle exceptions
- Connector-specific alternative to Try/Catch



Return Application Error Response stops the connector from creating a document error. Instead the connector returns an error message.

Technique:
Connector-Application Error Response





Technique: Retry

Types of Retry

- Retry in Try/Catch
- Process Retry Schedule
- Manual Document retry – Using original documents
- Process Design for Retry

Retry Considerations

- Process and documents have not changed, so will not solve all error types
- Remedies intermittent connectivity or availability errors in a connector
- Remedies intermittent lock-ups in target system

What Trigger setting is needed in the Try/Catch shape to allow a process level error to be caught and sent down the Catch path?

A | Document Errors

B | Process Errors

C | All Errors

This optional feature must be enabled within the Notify shape to have the message sent in a User Alert (Email or RSS).

A
Message Level must be set to Error

B
Enable Events

C
Enable User Log

D
Write Once per Execution

Assume that 5 Documents are being processed through the Try/Catch shape. If 2 of the 5 Documents fail, then which of the following happen?

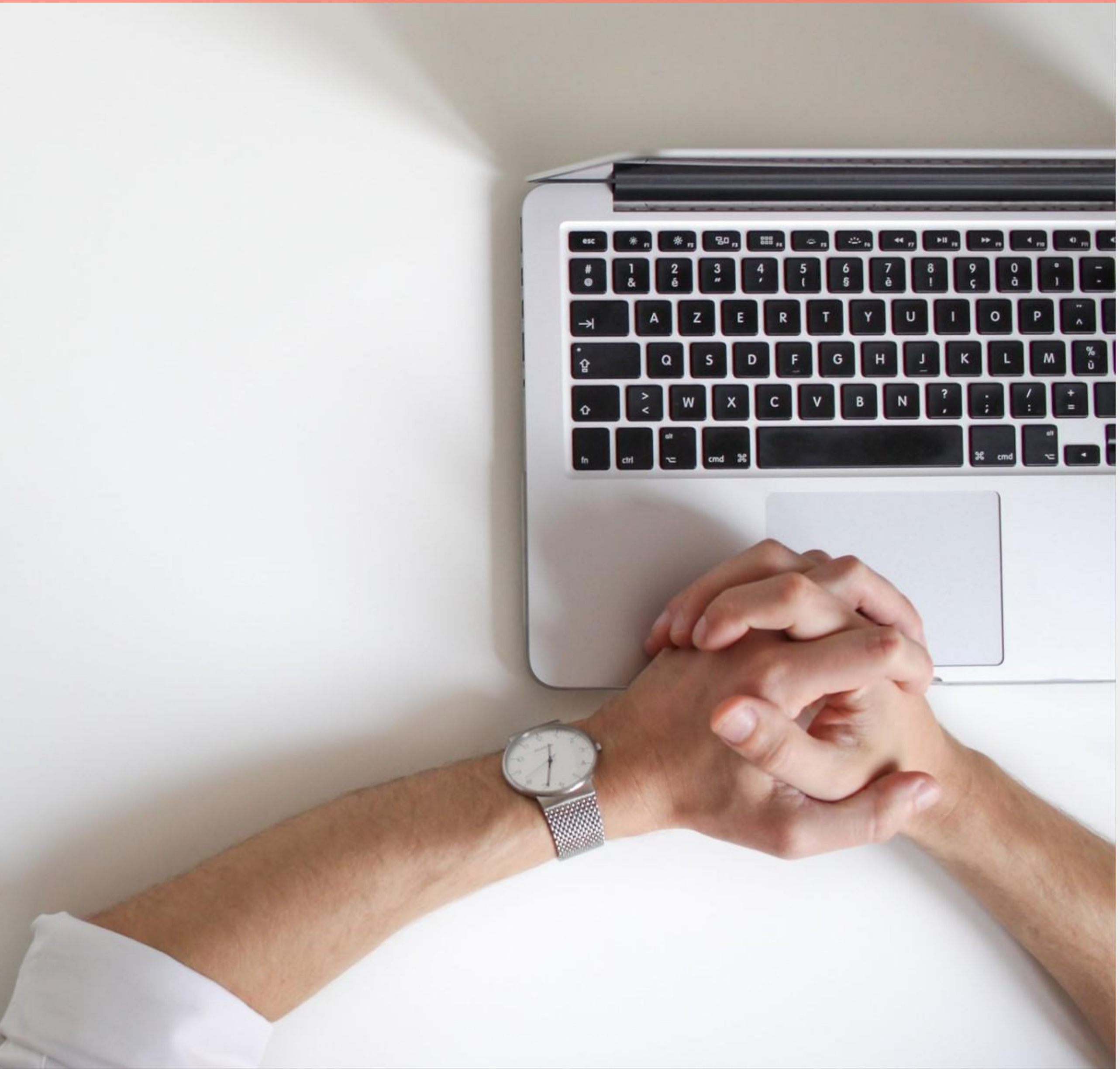
- **A** 1
- **B** 2
- **C** 3
- **D** 4
- **E** 5

A Flat File document flows through a Try/Catch shape down the Try path and is transformed in a Map shape to an XML document and renamed with a Set Properties shape, but fails at a Salesforce connector. When the document flows down the Catch path it will still be in XML format and renamed.

A. True

B. False

INSTRUCTOR TO DEMONSTRATE



Error Handling - Activity 2

Shared Web Server Event-Based Integration

What is Event-based Integration?

What is Event-based Integration?

- Recommended when processing is in response to irregularly occurring external events
- Typically, not executed on a schedule
- Instead, event-based processes listen for the occurrence of a specific event
- Web Service Publishing is one way to do this

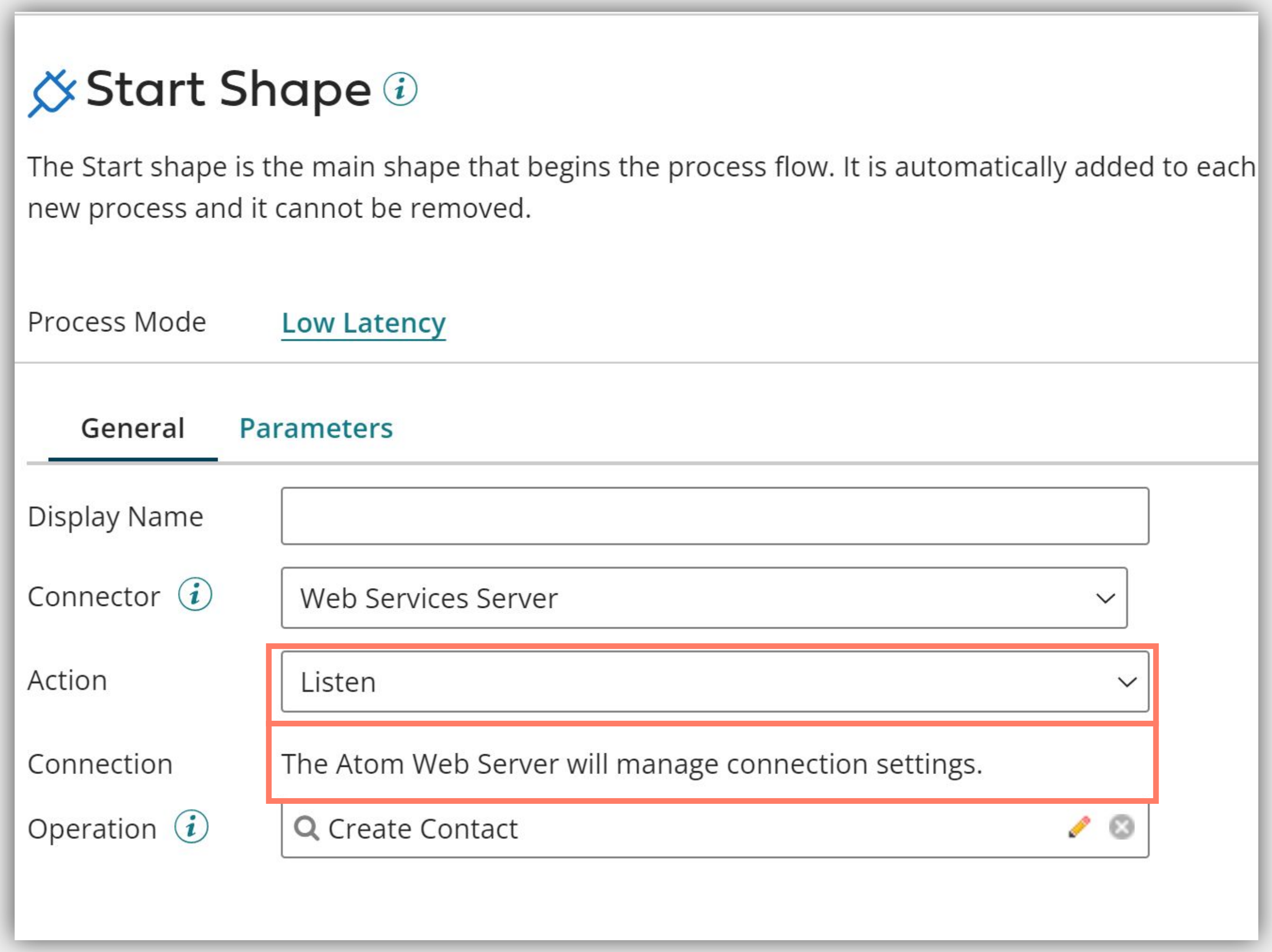
Web Service Publishing

AtomSphere allows you to:

- Turn any integration process into a web service.
- Deploy the web service on-premise or in the cloud.
- Use Atom's shared web-server to accept HTTP and SOAP requests in real time.
- Configure the Web server to control network/security settings.
- Publish a WSDL.

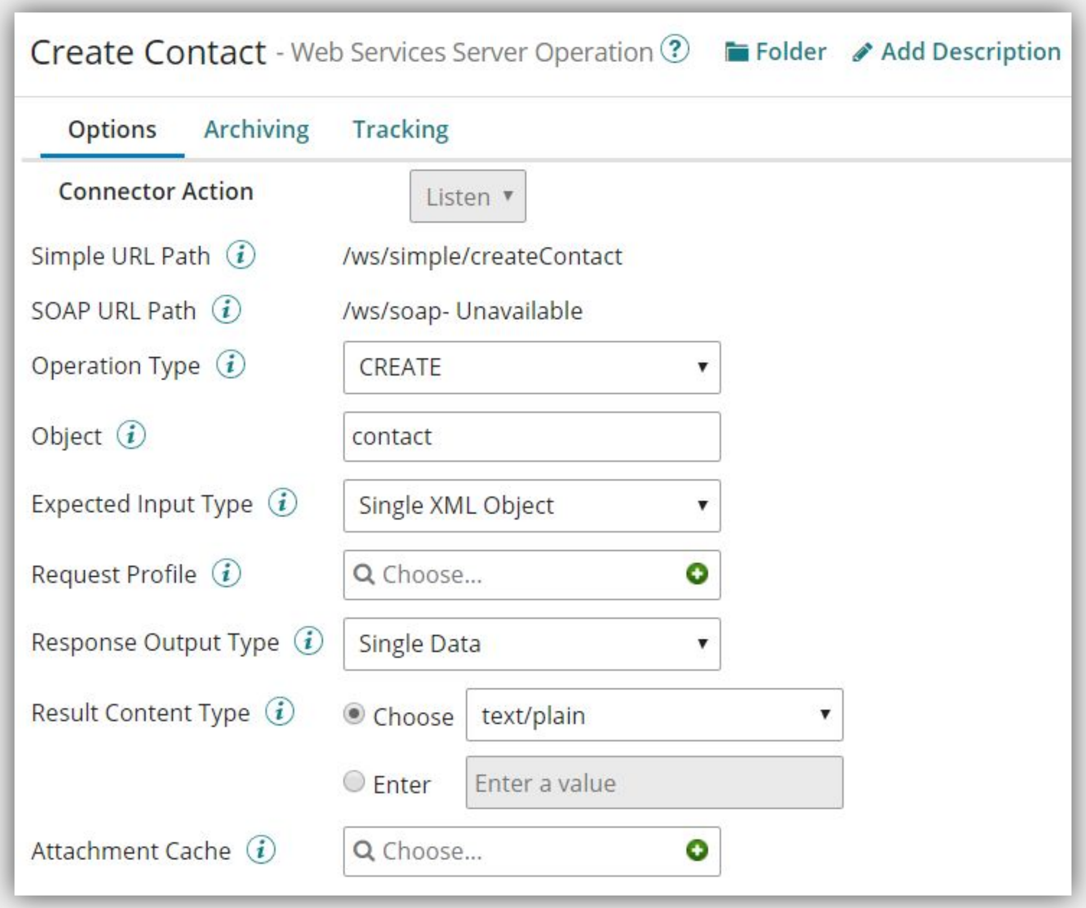
Web Services Server Connector

- Listen only connector
- Listens for & accepts REST, SOAP, and simple HTTP requests in real-time to initiate integration processes
- Simplest way to publish web services
- Connection managed by Atom Web Server



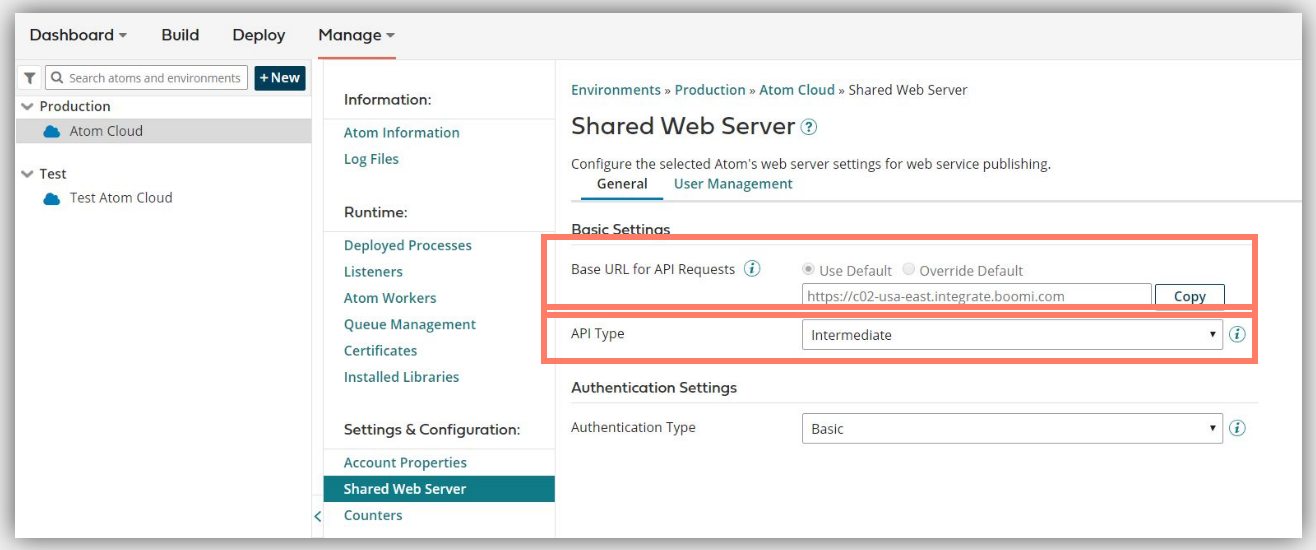
Web Services Server Operation

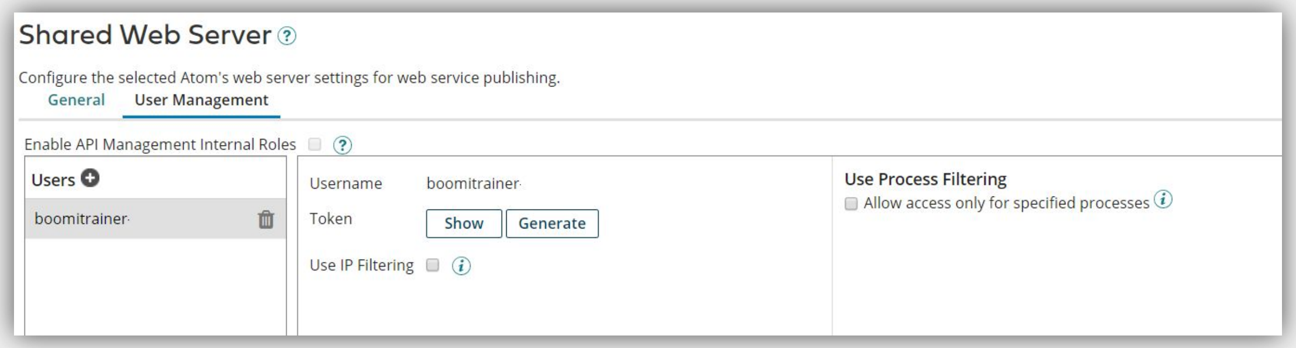
- Defines how to accept requests processed as standard documents.
- The example displays a Web Services Server Operation.
- The input and output types control the request and response data requirements for the process.
- Triggers and asynchronous processing are also configured on this tab.



Shared Web Server Settings and Configuration

- Shared server settings controlled at the Atom/Molecule/Cloud level and determine what type of requests can be made.
- Base URL for API Requests is found here.
- API Type is Intermediate when using the Web Services Server Connector, only specific users can hit the API.
- Available endpoints are /ws/simple/ and /ws/soap/
- Under User Management, Users can be added, Passwords generated, and Process Filtering enabled.

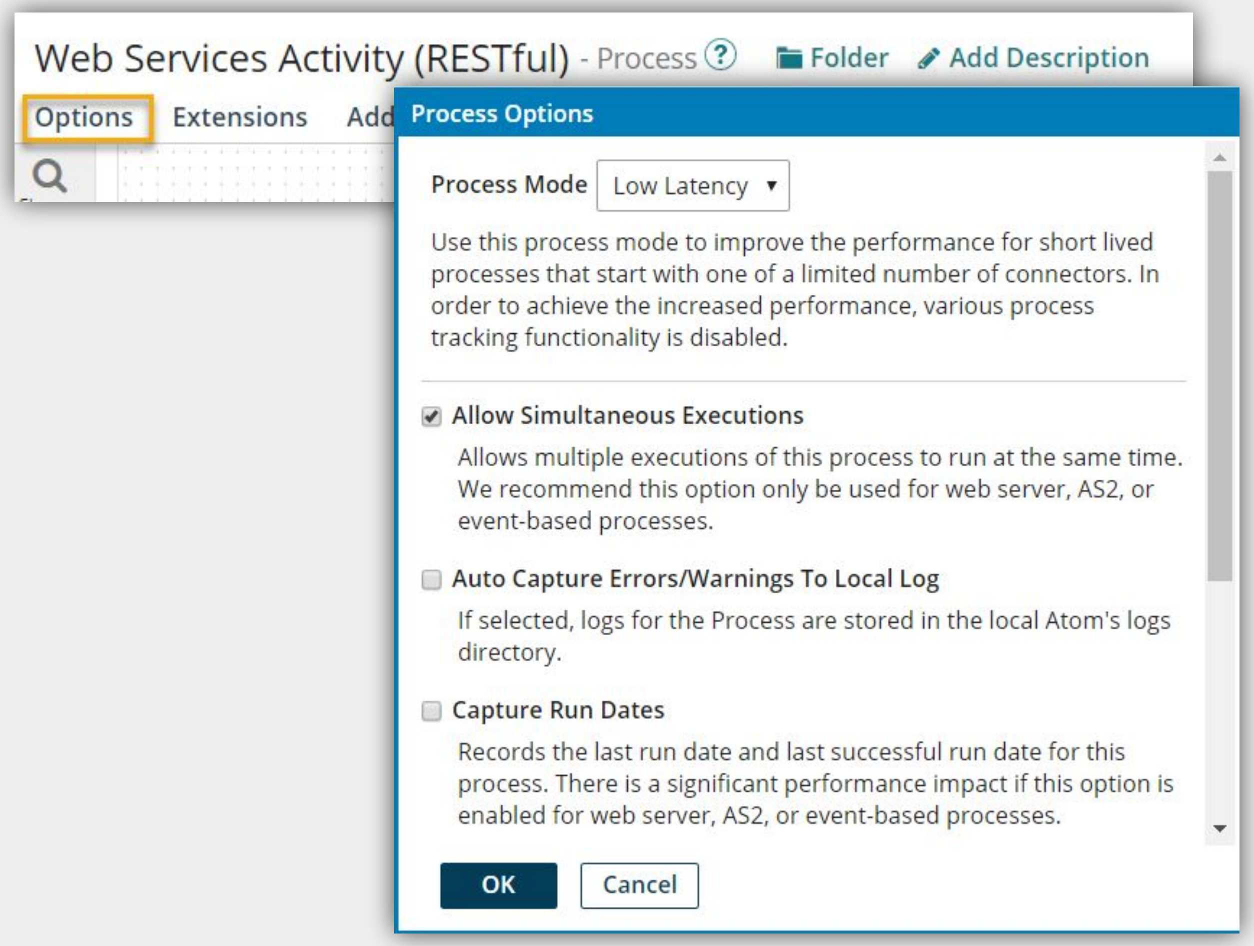




Support for Event-based Integration

Low Latency Process Option

- Request or response data is not recorded
- Process metrics are not recorded in the View Process State dialog, but the process log is available
- The inbound data's document logs are disabled

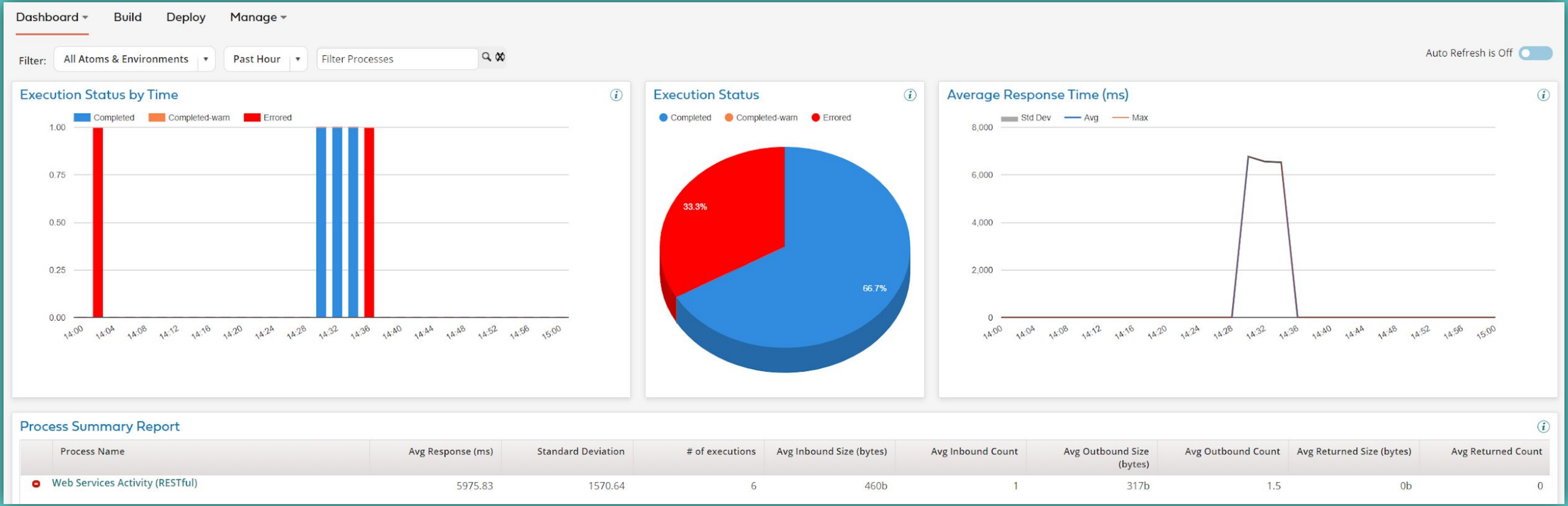


Web Services and an Atom Worker

- AtomSphere uses Atom workers to improve performance for certain types of requests in a Boomi or private Atom Cloud.
- A Web Service Server Connector is one connector that can use an Atom Worker.
- The maximum number of web service processes that can execute simultaneously is 20, with 10 queued.
- If using Low Latency mode, the maximum web-service process execution time is 30 seconds.

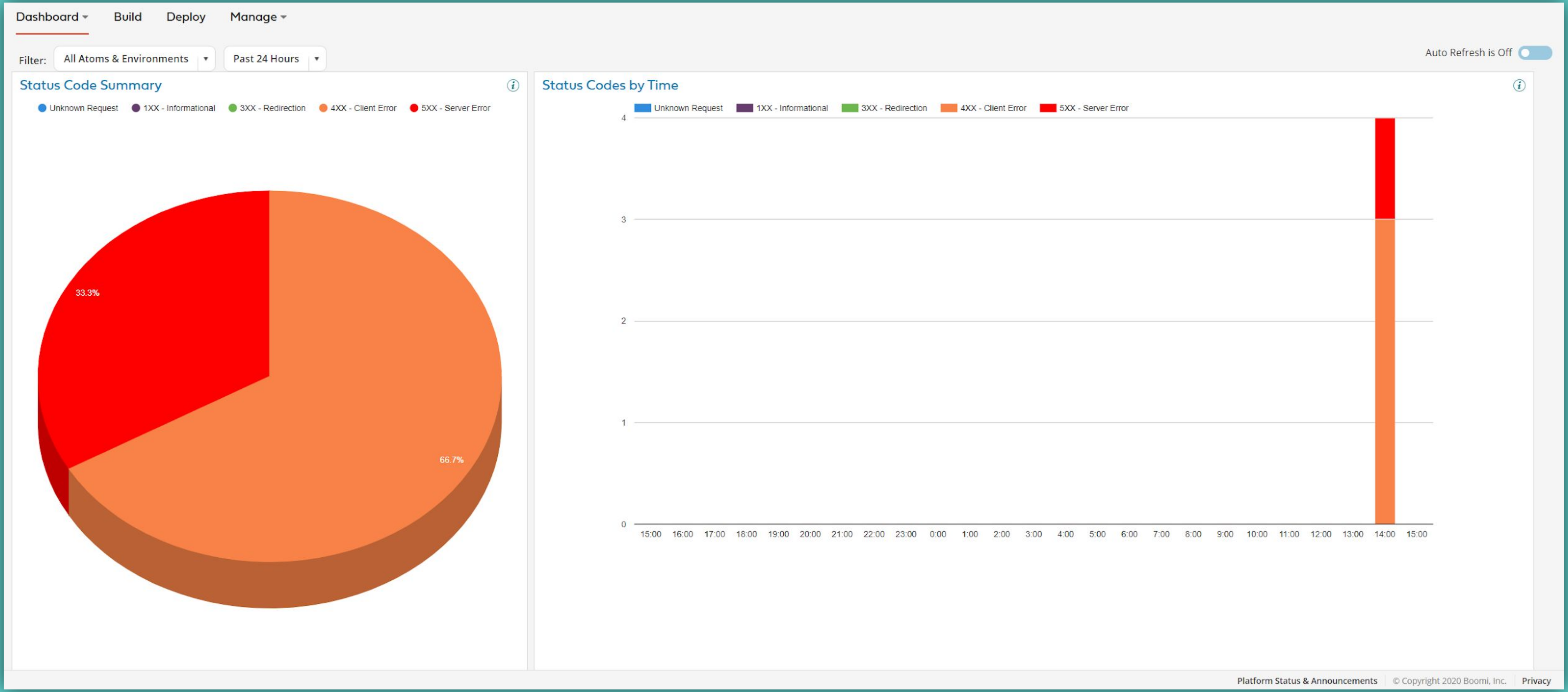
Real-time Dashboard

- Aggregated low latency process execution
- Provides data on execution status and average response time
- Does not include 400-level or 503 errors
- Summary data may be delayed by up to 5 minutes



HTTP Status Dashboard

- These logs are not available in Process Reporting or the Real-time Dashboard
- Codes are standard HTTP status codes, including 400-level and 503



If you execute a Low Latency process and the result is a 400-level error, the execution will appear as part of aggregate data in the HTTP Status Dashboard but not in the Real-time Dashboard.

A True

B False

When configuring the Web Services Server Connector, almost all of the configuration takes place in the:

A Connection

B Operation

AtomSphere has the ability to be a web service or to publish a WSDL

A. True

B. False

When using the Web Services Server Operation, the API Type needs to be set to which of the following:

A Basic

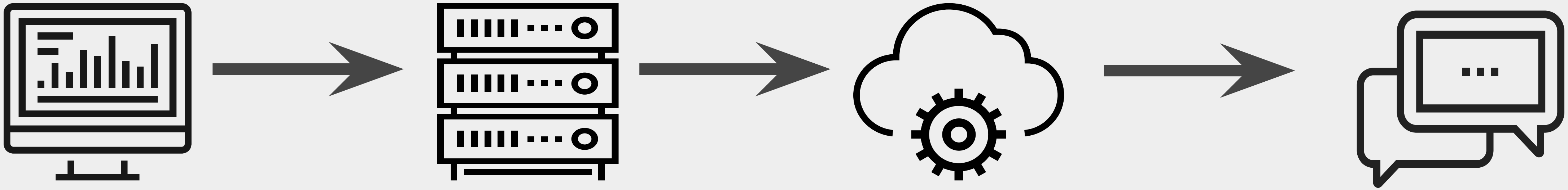
B Intermediate

C Advanced

D Expert

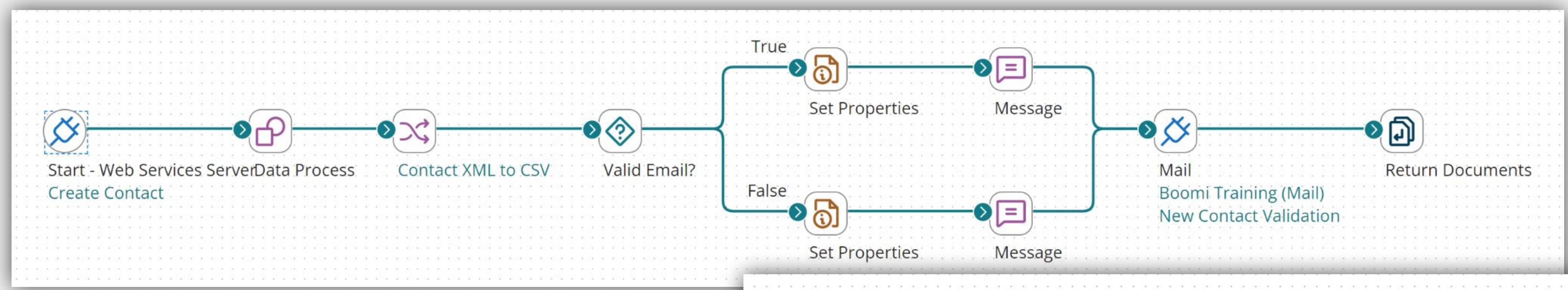
Business Use Case for Activity

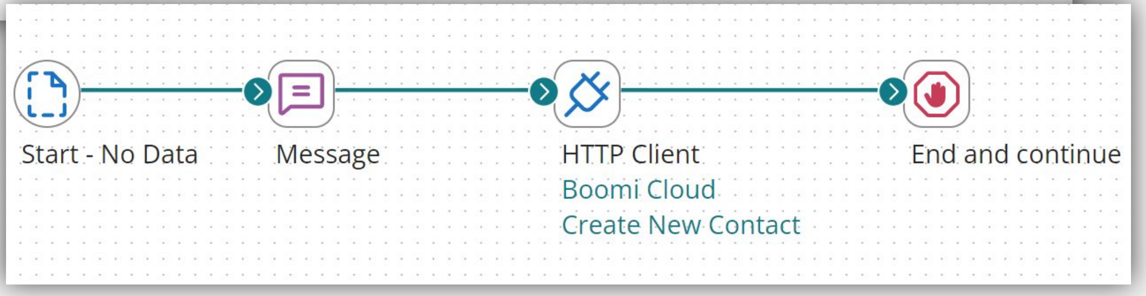
- Users enter contact information into an online form
- The application sends the information to the Boomi Web Server
- AtomSphere processes and validates the information and returns the appropriate responses
- Contact information is sent via email to the appropriate team member



Web Services Integration Activity

- Create a new Operation in the Web Services Server Connector to ‘Create Contact’
- Configure the Decision shape and Mail Operation, then add a Return Documents shape
- Package and Deploy the process
- Configure the Web Service Client to call the listener process
- Call the process, and confirm results





HANDS ON ACTIVITY

Web Services Integration Activity

THANK YOU.

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Publication Date	Summary
March 2020	Updated for Packaged Components


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